

# Prompt Engineering

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# 100 Real-World Prompt Engineering Exercises

## Exercise 1: Improve a Bad Prompt

### Situation

Many beginners write **very weak prompts**.

Example of a bad prompt:

Explain Python.

This prompt is bad because:

- No clarity
- No audience
- No output format
- No length limit

So AI will give **random answers**.

### Task

Improve this prompt using the **Prompt Formula**.

Prompt Formula:

Role + Task + Context + Constraints + Output Format

## Bad Prompt

Explain Python.

## Improved Prompt

You are a programming teacher.

Explain Python programming language for beginners.

Use simple English.

Give:

- 5 bullet points
- 1 real-life example
- 3 line summary

Keep explanation under 150 words.

## Why This Prompt is Better

It tells AI:

- **Role** → programming teacher
- **Audience** → beginners
- **Task** → explain Python
- **Constraints** → 150 words
- **Output format** → bullets + example + summary

So AI produces **controlled output**.

### **Your Practice Task**

Improve this bad prompt:

Tell me about machine learning.

Write a **better prompt**.

## Exercise 2: Role-Based Prompting

### Core Idea

AI gives **better answers when we assign a role**.

If we just ask a question, AI gives a **general answer**.

But when we give a **role**, AI responds like an expert.

### Example 1: Without Role

Prompt:

Explain how to save money.

AI may give a general answer.

### Example 2: With Role

Prompt:

You are a personal finance advisor.

Explain how a college student can save money every month.

Use simple English.

Give:

- 5 practical tips
- 1 real-life example

- 3 line summary

Now the answer will be **more practical and useful**.

## Why Role Prompting Works

Role gives **context** to the AI.

Examples of roles:

Teacher

Doctor

Lawyer

Software engineer

Marketing expert

Career coach

Business consultant

AI will **think from that role**.

## Real Life Example

Bad prompt:

Write interview questions.

Better prompt:

You are a senior HR manager.

Create 10 interview questions for a Python developer with 2 years experience.

## Your Practice Task

Convert this into a **role-based prompt**:

Explain data science.

Write a better prompt using a **role**.



## Exercise 3: Give Clear Context

### Core Idea

AI works better when it understands **the situation**.

If context is missing, AI gives **generic answers**.

Context means:

- who is asking
- why they are asking
- where it will be used

### Example 1: Without Context

Prompt:

Explain SQL.

AI will give a **general explanation**.

### Example 2: With Context

Prompt:

You are a database trainer.

Explain SQL for a student preparing for a software job interview.

Use simple English.

Include:

- 5 important SQL concepts
- 1 small example query
- 3 interview tips

Now AI understands:

- **role** → trainer
- **context** → job interview preparation
- **audience** → student

So the answer becomes **more useful**.

## Real-Life Example

Bad prompt:

Write a speech.

Better prompt:

You are a motivational speaker.

Write a 3-minute speech for college students who are worried about their career.

Use simple English and include one inspiring story.

### **Your Practice Task**

Improve this prompt by **adding context**.

Bad prompt:

Explain artificial intelligence.

Try to include:

- role
- audience
- situation

## Exercise 4: Specify the Output Format

### Core Idea

AI works much better when we **tell exactly how the answer should look**.

If we do not give a format, AI gives **long random paragraphs**.

But when we define format, the answer becomes **structured and clean**.

### Example 1: No Output Format

Prompt:

Explain machine learning.

AI will give long text.

### Example 2: With Output Format

Prompt:

You are a data science teacher.

Explain machine learning for beginners.

Use simple English.

Output format:

1. Definition (2 lines)
2. 5 Key Points
3. 1 Real-life Example
4. 3 Line Summary

Now AI produces **organized content**.

## **Why Output Format is Important**

It helps for:

- study notes
- blog writing
- teaching material
- social media posts
- interview preparation

## **Real-Life Example**

Bad prompt:

Write about leadership.

Better prompt:

You are a leadership coach.

Explain leadership for young professionals.

Output format:

1. Definition
2. 5 leadership qualities
3. 1 example
4. 3 line conclusion

### **Your Practice Task**

Improve this prompt by **adding output format**.

Bad prompt:

Explain cloud computing.

Try to include:

- role
- audience
- output format

## Exercise 5: Add Constraints to Control AI Output

### Core Idea

Constraints help us **control the answer**.

Without constraints, AI may give:

- very long answers
- complex language
- too many points

Constraints make the answer **precise and usable**.

### Example 1: No Constraints

Prompt:

Explain deep learning.

AI may give a **very long answer**.

### Example 2: With Constraints

Prompt:

You are an AI teacher.

Explain deep learning to a beginner student.

Constraints:

- Use simple English
- Maximum 120 words
- Give only 5 key points
- Include 1 small example

Output format:

1. Definition
2. 5 Key Points
3. Example
4. 3-line summary

Now AI response becomes **short, focused, and clear.**

## **Why Constraints Are Powerful**

Constraints help in:

- exam preparation
- study notes
- blog writing
- social media content
- interview answers

They **control the size and quality of output.**



## Real-Life Example

Bad prompt:

Write about time management.

Better prompt:

You are a productivity coach.

Explain time management for college students.

Constraints:

- Use simple English
- Maximum 100 words
- Give 4 practical tips

## Your Practice Task

Improve this prompt by **adding constraints**.

Bad prompt:

Explain blockchain.

Try to include:

- role
- audience
- constraints
- output format

## Exercise 6: Step-by-Step Prompting

### Core Idea

Sometimes we ask AI a **big question**.

AI may give **confusing answers**.

Better method is to ask AI to **think step by step**.

This improves:

- accuracy
- clarity
- reasoning

This technique is used in **Chain-of-Thought prompting**.

### Example 1: Normal Prompt

Solve this problem:

A shop sells a pen for ₹20.

Rahul buys 7 pens.

How much money did he pay?

AI might answer directly.

**Example 2: Step-by-Step Prompt**

You are a math teacher.

Solve the problem step by step.

Problem:

A pen costs ₹20.

Rahul buys 7 pens.

Instructions:

1. First calculate cost of one pen
2. Then multiply with number of pens
3. Show the final answer clearly

Now AI will show:

Step 1

Step 2

Step 3

Final answer

## Why Step-by-Step Prompting is Powerful

It is useful for:

- math problems
- coding problems
- logical reasoning
- debugging programs
- interview preparation

### Real-Life Example

Bad prompt:

Write Python code for sorting numbers.

Better prompt:

You are a Python teacher.

Explain how to sort numbers in Python.

Instructions:

1. Explain the concept
2. Show Python code
3. Explain each line of code
4. Give one example input and output

## Your Practice Task

Improve this prompt using **step-by-step instructions**.

Bad prompt:

Explain how to build a website.

Try to include:

- role
- step-by-step explanation
- simple English

## Exercise 7: Few-Shot Prompting

### Core Idea

Few-shot prompting means **giving examples to the AI**.

When AI sees examples, it **learns the pattern** and gives better answers.

This is very useful for:

- formatting output
- classification tasks
- training AI behaviour
- automation systems

### Example 1: Without Examples

Prompt:

Classify this sentence as Positive or Negative.

Sentence:

The movie was amazing.

AI may respond correctly, but sometimes answers may be inconsistent.

## Example 2: Few-Shot Prompt

You are a sentiment analyzer.

Classify the sentence as **Positive or Negative**.

Examples:

Sentence: I love this product

Answer: Positive

Sentence: This service is terrible

Answer: Negative

Sentence: The movie was amazing

Answer:

Now AI understands the **pattern**.

Expected output:

Positive

## Why Few-Shot Prompting is Powerful

It is used in:

- AI chatbots
- email classification
- resume screening
- customer support automation

- sentiment analysis

Many AI tools use **this technique internally**.

## Real-Life Example

Prompt:

You are an interview evaluator.

Examples:

Answer: I don't know anything about Python

Rating: Poor

Answer: I know Python basics like variables, loops, and functions

Rating: Average

Answer: I built Python projects using APIs and databases

Rating: Excellent

Candidate Answer:

I created automation scripts using Python.

Rating:

AI will follow the **same evaluation pattern**.



## Your Practice Task

Improve this prompt using **Few-Shot examples**.

Bad prompt:

Identify whether the email is Spam or Not Spam.

Try to include:

- role
- 2 examples
- classification pattern

## Exercise 8: Prompt Chaining

### Core Idea

Prompt chaining means **using multiple prompts step-by-step**.

Output of **Prompt 1** → becomes input of **Prompt 2**.

This helps to solve **complex tasks**.

Many AI applications use this method.

### Example: Blog Writing System

Instead of one big prompt, we break it into steps.

#### **Prompt 1: Generate Ideas**

You are a content strategist.

Generate 5 blog topic ideas about Artificial Intelligence for beginners.

**Prompt 2: Create Outline**

Take the following topic and create a blog outline.

Topic: {{Topic\_from\_prompt1}}

Output format:

- Introduction
- 5 sections
- Conclusion

**Prompt 3: Write Blog**

Write a beginner-friendly blog using this outline.

Outline: {{Outline\_from\_prompt2}}

Constraints:

- Use simple English
- Maximum 500 words

Now the workflow becomes:

Idea → Outline → Blog

This is **Prompt Chaining**.

## Why Prompt Chaining is Powerful

It is used in:

- AI content generators
- resume analyzers
- interview preparation systems
- study note generators
- chatbots

## Real-Life Example

AI Study Tool Workflow:

Prompt 1 → Explain topic

Prompt 2 → Generate summary

Prompt 3 → Generate MCQs

You actually used this system earlier.

## Your Practice Task

Design a **prompt chain for resume analysis**.

Example workflow:

Step 1 → Analyze resume

Step 2 → Find weaknesses

Step 3 → Generate interview questions

Write **3 prompts for this chain**.

Reply with your prompts.

## Exercise 9: Persona Prompting

### Core Idea

Persona prompting means **asking AI to behave like a specific person or expert.**

When AI acts as a persona, the answers become **more realistic and specialized.**

Persona can be:

- teacher
- CEO
- doctor
- HR manager
- startup mentor
- marketing expert
- software architect

### Example 1: Normal Prompt

How can a startup grow?

AI gives a **general answer.**

## Example 2: Persona Prompt

You are a startup mentor with 20 years of experience.

Advise a new startup founder on how to grow their business in the first year.

Use practical strategies.

Output format:

1. 5 growth strategies
2. 1 real-life example
3. 3 mistakes to avoid

Now AI gives **expert-style advice**.

## Why Persona Prompting Works

Persona adds:

- expertise
- tone
- perspective
- domain knowledge

This technique is used in:

- consulting bots

- education tools
- coaching assistants
- business advisors

## Real-Life Example

Prompt:

You are a senior software architect.

Explain how to design a scalable web application for 1 million users.

Use simple explanation for junior developers.

## Your Practice Task

Improve this prompt using **persona prompting**.

Bad prompt:

Explain how to prepare for a job interview.

Try to include:

- persona
- audience
- output format

## Exercise 10: Prompt Refinement

### Core Idea

Prompt refinement means **improving a prompt again and again** until the result becomes perfect.

Professional prompt engineers **do not write one prompt**. They **test → improve → test again**.

This process is called **prompt iteration**.

### Example: First Prompt (Weak)

Explain data analytics.

Problem:

- Too vague
- No audience
- No format
- No constraints



## **Improved Prompt (Version 2)**

You are a data analyst.

Explain data analytics for beginners.

Use simple English.

Give:

- 5 key concepts
- 1 real-life example
- 3 line summary

Better, but still missing clarity.

## **Final Prompt (Version 3)**

You are a senior data analyst.

Explain data analytics to college students who have no technical background.

Constraints:

- Use very simple English
- Maximum 120 words

Output format:

1. Definition (2 lines)
2. 5 Key Concepts
3. Real-life Example
4. 3-line Summary

Now the prompt is **clear, structured, and controlled**.

## Why Prompt Refinement is Important

This is used in:

- AI product development
- chatbot design
- AI automation systems
- prompt engineering jobs

Good prompts come from **continuous improvement**.

## Real-Life Example

Prompt 1:

Write a marketing email.

Prompt 2:

Write a marketing email for a new online course.

Prompt 3 (Refined):

You are a digital marketing expert.

Write a persuasive email promoting a new Python course for beginners.

Constraints:

- Maximum 120 words
- Friendly tone

Output format:

- Subject line
- Email body
- Call to action

## Your Practice Task

Refine this prompt into a **strong prompt**.

Bad prompt:

Write about artificial intelligence.

Try to include:

- role
- audience
- constraints
- output format

## Exercise 11: Hallucination Control

### Core Idea

Sometimes AI **creates information that is not true**.  
This is called **hallucination**.

Example:

AI may invent:

- fake facts
- wrong data
- non-existing sources

Prompt engineers must **control hallucination**.

### Example: Normal Prompt

Who invented Python?

AI answer:

Guido van Rossum.

This is correct.

But sometimes if AI is unsure, it may generate **wrong information**.

## **Better Prompt to Reduce Hallucination**

You are a technology historian.

Answer the question only if you are confident.

If the information is uncertain, say:

"I am not sure about this."

Question:

Who invented Python?

Now AI becomes **more careful**.

## **Another Example**

Bad prompt:

List 10 research papers on quantum AI.

AI might invent fake papers.

Better prompt:

List well-known research papers on quantum computing.

If you are unsure about any paper, say "uncertain".

Do not invent titles.

## Why Hallucination Control is Important

It is critical in:

- medical AI
- legal AI
- research tools
- finance applications

Incorrect information can be **dangerous**.

## Real-Life Example

Prompt:

You are a medical assistant.

Provide health advice only based on common medical knowledge.

If you are unsure, recommend consulting a doctor.

## Your Practice Task

Improve this prompt to **avoid hallucination**.

Bad prompt:

Give statistics about AI usage in the world.

Try to include:

- instruction not to guess
- instruction to say "not sure" if data is uncertain

## Exercise 12: Prompt Templates

### Core Idea

Prompt templates are **reusable prompts**.

Instead of writing a new prompt every time, we create a **template with variables**.

Example variables:

{topic}

{audience}

{word\_limit}

This helps build **AI tools and automation systems**.

### Example: Normal Prompt

Explain Python for beginners.

This works only for **Python**.



## Prompt Template

You are a teacher.

Explain **{topic}** for **{audience}**.

Constraints:

- Use simple English
- Maximum **{word\_limit}** words

Output format:

1. Definition
2. 5 Key Points
3. Example
4. Summary

Now we can reuse it.

Example 1

Topic = Python

Audience = beginners

Word limit = 120

Example 2

Topic = Machine Learning

Audience = college students

Word limit = 150

Same template works.

## Why Prompt Templates Are Powerful

They are used in:

- AI apps
- chatbots
- study tools
- content generators
- resume analyzers

Many AI products run on **prompt templates**.

## Real-Life Example

AI Study Notes Generator Template:

You are a study notes generator.

Explain **{topic}** for **{target\_audience}**.

Output format:

1. Topic
2. 7 Key Notes

3. Real-life Example

4. Quick Summary

### Your Practice Task

Create a **prompt template** for this task:

Generating **interview questions**.

Your template should include variables like:

- {job\_role}
- {experience\_level}
- {number\_of\_questions}

## Exercise 13: Prompt Evaluation

### Core Idea

Writing prompts is not enough.

We must also **test and evaluate them**.

Prompt evaluation means checking:

- Is the answer correct?
- Is the format followed?
- Is the language simple?
- Is the response consistent?

This helps us **improve prompt quality**.

### Example Prompt

You are a programming teacher.

Explain Python loops for beginners.

Constraints:

- Use simple English
- Maximum 120 words

Output format:

1. Definition
2. 3 Types of loops
3. Example
4. Summary

## **How to Evaluate the Output**

Check these things:

### **1 Correctness**

Is the explanation correct?

### **2 Format**

Did AI follow the format?

### **3 Clarity**

Is it easy to understand?

### **4 Length**

Is it within the word limit?

### **5 Consistency**

If we run again, do we get similar answers?

## Prompt Evaluation Checklist

Good prompts should have:

- ✓ Clear instructions
- ✓ Defined format
- ✓ Controlled length
- ✓ Correct information
- ✓ Consistent responses

## Real-Life Example

AI Resume Analyzer Prompt

Evaluation questions:

- Did AI correctly identify skills?
- Did AI generate relevant interview questions?
- Did AI follow output format?

If not → **prompt needs improvement.**

## Your Practice Task

You created a prompt to:

Generate interview questions.

Now evaluate it using these **4 criteria**:

1 Correctness

2 Clarity

3 Format

4 Usefulness

## Exercise 14: Prompt Debugging

### Core Idea

Sometimes prompt does not give expected output.

Instead of blaming AI, we must **debug the prompt**.

Prompt debugging means:

- Find what is missing
- Find what is unclear
- Improve instructions

### Example: Problem Prompt

You are a teacher.

Explain Java.

Output format:

1. Definition
2. Example

Problem:

- No audience
- No length control



- No clarity level

AI answer may be too complex.

### **Debugged Prompt**

You are a programming teacher.

Explain Java to a 1st year engineering student.

Constraints:

- Use simple English
- Maximum 120 words

Output format:

1. Definition (2 lines)
2. 4 Key Features
3. One simple example
4. 3-line summary

Now the prompt is fixed.

## Common Prompt Errors

- 1 Too vague
- 2 No audience defined
- 3 No output format
- 4 No constraints
- 5 Too many instructions mixed together

## Real-Life Example

Bad prompt:

Write a business plan.

Debugged prompt:

You are a startup consultant.

Create a simple business plan for a small tea shop in Hyderabad.

Constraints:

- Maximum 300 words
- Include cost estimate
- Include marketing strategy

Output format:

1. Idea
2. Target customers

3. Budget
4. Marketing plan
5. Expected profit

### **Your Practice Task**

Debug this prompt:

Write about cybersecurity.

Improve it by adding:

- role
- audience
- constraints
- output format

## Exercise 15: Tone Control

### Core Idea

Tone control means **deciding how AI should speak**.

Same content can be written in **different tones**.

Examples of tone:

- Friendly
- Professional
- Motivational
- Funny
- Formal
- Simple teaching style

Prompt engineers use tone to **control the personality of AI responses**.

### Example Topic

Time Management

### **Prompt 1: Professional Tone**

You are a productivity expert.

Explain time management for working professionals.

Tone: Professional

Output:

- 5 practical tips
- Short explanation

### **Prompt 2: Friendly Tone**

You are a friendly mentor.

Explain time management to a college student.

Tone: Friendly and simple.

### **Prompt 3: Motivational Tone**

You are a motivational speaker.

Explain why time management is important for success.

Tone: Inspirational and energetic.

Same topic → **different tone** → **different output style**.

## **Why Tone Control Is Important**

It is used in:

- marketing content
- education content
- social media posts
- business communication
- storytelling

Tone changes **how people feel about the message**.

## **Real-Life Example**

Bad prompt:

Explain leadership.

Better prompt:

You are a leadership coach.

Explain leadership to young entrepreneurs.

Tone: Motivational and inspiring.

Output:

- 5 leadership lessons
- 1 inspiring story

## Your Practice Task

Improve this prompt by **controlling the tone**.

Bad prompt:

Explain the importance of learning programming.

Try to include:

- role
- audience
- tone
- output format

## Exercise 16: Length Control

### Core Idea

Sometimes we need **short answers**.

Sometimes we need **detailed explanations**.

Prompt engineers control the **length of AI output**.

Without length control, AI may give:

- very long answers
- too much information

Length control helps create:

- study notes
- social media posts
- summaries
- quick explanations

### Example Topic

Artificial Intelligence



### **Prompt 1: Short Answer**

You are an AI teacher.

Explain artificial intelligence for beginners.

Constraints:

- Maximum **50 words**

### **Prompt 2: Medium Explanation**

You are an AI teacher.

Explain artificial intelligence for beginners.

Constraints:

- Maximum **150 words**

### **Prompt 3: Detailed Explanation**

You are an AI teacher.

Explain artificial intelligence.

Constraints:

- Maximum **500 words**
- Include examples

Same topic → **different length** → **different depth of explanation.**

## **Why Length Control is Important**

It is useful for:

- Instagram captions
- YouTube scripts
- study notes
- interview answers
- blog writing

## **Real-Life Example**

Bad prompt:

Explain machine learning.

Better prompt:

You are a data science teacher.

Explain machine learning to beginners.

Constraints:

- Maximum **100 words**

Output format:

1. Definition
2. 3 Key Points
3. Example

## Your Practice Task

Improve this prompt using **length control**.

Bad prompt:

Explain blockchain technology.

Try to include:

- role
- audience
- word limit
- output format

## Exercise 17: Prompt for Content Creation

### Core Idea

Prompt engineering is widely used for **content generation**.

AI can create:

- blog posts
- YouTube scripts
- social media posts
- study notes
- marketing content

But the prompt must clearly define **structure and purpose**.

### Example: Blog Writing Prompt

You are a technology writer.

Write a beginner-friendly blog about Artificial Intelligence.

Constraints:

- Use simple English
- Maximum 400 words

Output format:

1. Introduction
2. What is AI
3. Applications of AI
4. Benefits
5. Conclusion

### **Example: Social Media Prompt**

You are a social media content creator.

Create an Instagram post about the benefits of learning Python.

Constraints:

- Maximum 80 words
- Friendly tone

Output format:

1. Hook
2. 3 benefits
3. Call to action

### **Example: YouTube Script Prompt**

You are a YouTube educator.

Write a short script explaining Machine Learning.

Constraints:

- Maximum 1 minute script
- Simple language

Output format:

1. Hook
2. Explanation
3. Example
4. Closing line

### **Why Content Prompts Are Important**

These prompts are used in:

- AI writing tools
- marketing automation
- educational platforms
- content creation systems

## Real-Life Example

Prompt:

You are a career mentor.

Create a LinkedIn post about the importance of learning AI skills.

Constraints:

- Maximum 120 words
- Professional tone

Output format:

- Opening line
- 3 key points
- Closing message

## Your Practice Task

Improve this prompt for **content creation**.

Bad prompt:

Write about the benefits of learning AI.

Try to include:

- role

- audience
- content type
- constraints
- output format



## Exercise 18: Prompt for Code Generation

### Core Idea

AI can generate **programming code**, but the prompt must clearly specify:

- programming language
- task
- input and output
- explanation requirement

Without these, AI may generate **incorrect or incomplete code**.

### Example 1: Weak Prompt

Write code for sorting numbers.

Problems:

- No language specified
- No explanation
- No example input/output

## Improved Prompt

You are a Python programming teacher.

Write a Python program to sort a list of numbers in ascending order.

Constraints:

- Use simple code suitable for beginners
- Add comments in the code

Output format:

1. Python code
2. Explanation of each step
3. Example input and output

## Why This Prompt Works Better

It clearly tells AI:

- language → Python
- difficulty level → beginner
- explanation needed
- example required

## Real-Life Example

Prompt:

You are a software developer.

Write a Python program that checks whether a number is prime.

Constraints:

- Code should be beginner friendly
- Include comments

Output format:

1. Python code
2. Explanation
3. Example run

## Where This Prompt Type Is Used

- coding assistants
- AI programming tools
- coding education platforms
- interview preparation tools

Many developers use AI **daily for coding tasks**.

## **Your Practice Task**

Improve this prompt:

Write code to create a login system.

Try to include:

- programming language
- role
- explanation requirement
- output format

## Exercise 19: Prompt for Data Analysis

### Core Idea

Prompt engineering is also used for **data analysis**.

AI can help to:

- analyze data
- find patterns
- summarize insights
- create reports

But the prompt must clearly mention:

- what the data is
- what analysis is required
- what output format is needed

### Example 1: Weak Prompt

Analyze this data.

Problem:

- No context

- No objective
- No format

AI may give random analysis.

### **Improved Prompt**

You are a data analyst.

Analyze the following sales data.

Data:

January: ₹50,000

February: ₹65,000

March: ₹80,000

April: ₹70,000

Tasks:

1. Identify the sales trend
2. Find the highest sales month
3. Give 2 business insights

Output format:

1. Trend analysis
2. Key observation
3. Business insights

## Why This Prompt Works

It clearly defines:

- role → data analyst
- data → provided values
- task → analyze trend
- format → structured output

## Real-Life Example

Prompt:

You are a business analyst.

Analyze the following customer feedback data.

Tasks:

- identify common problems
- suggest improvements

Output format:

1. Main issues
2. Possible reasons
3. Suggested improvements

## **Where This Is Used**

- business analytics
- marketing analysis
- customer feedback analysis
- financial reporting
- product improvement

## **Your Practice Task**

Improve this prompt:

Analyze student performance data.

Try to include:

- role
- data example
- analysis task
- output format



## Exercise 20: Prompt for Decision Making

### Core Idea

Prompt engineering can help in **decision making**.

AI can help to:

- compare options
- evaluate pros and cons
- suggest better choices

But the prompt must clearly specify:

- the decision problem
- available options
- evaluation criteria

### Example 1: Weak Prompt

Which business is better?

Problem:

- No options
- No criteria

- No context

AI cannot give useful advice.

### **Improved Prompt**

You are a business consultant.

Help me decide which business is better.

Options:

1. Open a coffee shop
2. Start an online clothing store

Evaluation criteria:

- investment required
- profit potential
- risk level

Output format:

1. Comparison table
2. Pros and cons of each option
3. Final recommendation

## Why This Prompt Works

It clearly defines:

- role → business consultant
- options → two businesses
- criteria → investment, profit, risk
- format → comparison + recommendation

## Real-Life Example

Prompt:

You are a career advisor.

Help a student decide between two careers.

Options:

1. Software engineer
2. Data analyst

Evaluation criteria:

- job demand
- salary potential
- learning difficulty

Output format:

1. Comparison
2. Advantages and disadvantages
3. Final suggestion

### **Your Practice Task**

Improve this prompt:

Which programming language should I learn?

Try to include:

- role
- options
- evaluation criteria
- output format

## Exercise 21: Prompt for Summarization

### Core Idea

Sometimes we have **long text** like:

- articles
- research papers
- meeting notes
- books

Prompt engineering helps AI **convert long text into short summaries**.

### Example 1: Weak Prompt

Summarize this text.

Problem:

- No length limit
- No structure
- No audience

## Improved Prompt

You are a study assistant.

Summarize the following article.

Constraints:

- Use simple English
- Maximum 80 words

Output format:

1. Main idea
2. 5 key points
3. 2-line conclusion

## Why This Prompt Works

It clearly defines:

- role → study assistant
- length → 80 words
- format → structured summary

This makes summaries **clean and useful**.

## **Real-Life Example**

Prompt:

You are a meeting assistant.

Summarize the following meeting discussion.

Constraints:

- Maximum 5 bullet points

Output format:

1. Key decisions
2. Action items
3. Important notes

## **Where This Is Used**

- research summaries
- book summaries
- meeting notes
- study materials
- news summarization

## **Your Practice Task**

Improve this prompt:

Summarize this blog post.

Try to include:

- role
- constraints
- output format



## Exercise 22: Prompt for Question Generation

### Core Idea

AI can generate **questions from content**.

This is widely used in:

- online learning platforms
- exam preparation tools
- quiz generators
- interview preparation systems

Prompt must clearly specify:

- type of questions
- difficulty level
- number of questions
- format

**Example 1: Weak Prompt**

Create questions from this text.

Problem:

- No question type
- No difficulty level
- No format

**Improved Prompt**

You are an exam paper setter.

Generate questions from the following text.

Constraints:

- Create 5 multiple-choice questions
- Suitable for beginners

Output format:

Question

A) Option

B) Option

C) Option

D) Option

Answer

## Why This Prompt Works

It clearly defines:

- role → exam paper setter
- number → 5 questions
- type → MCQ
- format → structured output

## Real-Life Example

Prompt:

You are a teacher.

Generate 5 interview questions from the following Python topic.

Constraints:

- Beginner level
- Include answers

Output format:

1. Question
2. Answer

## **Where This Is Used**

- learning apps
- AI tutors
- interview preparation tools
- exam practice platforms

## **Your Practice Task**

Improve this prompt:

Generate questions from this topic.

Try to include:

- role
- question type
- number of questions
- output format

## Exercise 23: Prompt for Classification

### Core Idea

Classification means **sorting information into categories**.

AI can classify:

- emails → Spam / Not Spam
- reviews → Positive / Negative
- resumes → Fresher / Experienced
- support tickets → Technical / Billing / General

Prompt must clearly define:

- categories
- input format
- output format

### Example 1: Weak Prompt

Classify this review.

Problem:

- No categories defined

- No format

### Improved Prompt

You are a sentiment analyzer.

Classify the following review as **Positive, Negative, or Neutral**.

Review:

"The product quality is excellent and delivery was fast."

Output format:

Sentiment: \_\_\_\_\_

### Why This Prompt Works

It clearly defines:

- role → sentiment analyzer
- categories → Positive / Negative / Neutral
- format → structured output

### Real-Life Example

Prompt:

You are a resume screening assistant.

Classify the candidate as **Fresher or Experienced**.

Input:

Resume text: {{resume}}

Rules:

- If experience is 0 years → Fresher
- If experience is 1+ years → Experienced

Output format:

Candidate Type: \_\_\_\_\_

### **Where Classification Prompts Are Used**

- HR automation
- email filtering
- customer support systems
- survey analysis
- sentiment analysis

### **Your Practice Task**

Improve this prompt:

Classify this email.

Try to include:

- role

- categories
- output format

## Exercise 24: Prompt for Information Extraction

### Core Idea

Information extraction means **pulling specific data from text**.

AI can extract:

- name
- email
- phone number
- skills
- dates
- locations

This technique is widely used in:

- resume analyzers
- form processing
- document analysis
- CRM automation



### **Example 1: Weak Prompt**

Extract information from this text.

Problem:

- Not clear what information is needed
- No format defined

### **Improved Prompt**

You are a data extraction assistant.

Extract the following information from the text.

Fields to extract:

- Name
- Email
- Phone number
- Location

Text:

Rahul Kumar

Email: rahul123@gmail.com

Phone: 9999999999

Location: Hyderabad

Output format:

Name:

Email:

Phone:

Location:

## Why This Prompt Works

It clearly defines:

- role → extraction assistant
- fields → name, email, phone
- format → structured output

This makes the result **clean and usable**.

## Real-Life Example

Prompt:

You are a resume analyzer.

Extract the following details from the resume.

Fields:

- Candidate name
- Skills
- Education

- Experience years

Output format:

Name:

Skills:

Education:

Experience:

### **Where This Is Used**

- HR resume screening
- document automation
- data entry automation
- CRM systems
- chatbot form processing

### **Your Practice Task**

Improve this prompt:

Extract details from this resume.

Try to include:

- role
- fields to extract
- output format

## Exercise 25: Prompt for AI Agents

### Core Idea

An **AI Agent** is an AI that can:

1. Think
2. Take action
3. Observe results
4. Decide the next step

This process is called **Reason → Action → Observation (ReAct)**.

Prompt engineers design prompts that guide AI through these steps.

### Example: AI Research Assistant

Prompt:

You are an AI research assistant.

Your task is to answer the user's question by following these steps.

Steps:

1. **Reason** – Think about the question carefully
2. **Action** – Search or analyze the information
3. **Observation** – Review the result
4. **Final Answer** – Give the best answer

Question:

What are the main benefits of learning Python?

Output format:

Reason:

Action:

Observation:

Final Answer:

## Why This Prompt Works

It forces AI to:

- think logically
- avoid guessing
- show reasoning
- produce better answers

This method is widely used in **AI agents and automation tools**.

## **Real-Life Example**

Prompt:

You are an AI troubleshooting assistant.

Follow this process:

1. Understand the problem
2. Suggest possible solutions
3. Evaluate each solution
4. Provide the best recommendation

Problem:

My Python code is giving an error.

Output format:

Problem Understanding:

Possible Solutions:

Evaluation:

Best Solution:

## **Where AI Agent Prompts Are Used**

- AI assistants
- coding copilots

- research assistants
- automation systems
- smart chatbots

## Your Practice Task

Improve this prompt to create an **AI learning assistant agent**.

Bad prompt:

Help students learn programming.

Try to include:

- role
- step-by-step reasoning
- structured output

## Exercise 26: Prompt for Multi-Step Problem Solving

### Core Idea

Some problems cannot be solved in **one step**.

We must break them into **multiple logical steps**.

Prompt engineers guide AI to **solve problems step-by-step**.

This improves:

- reasoning
- accuracy
- clarity

### Example Problem

A small shop owner wants to increase sales.

Weak prompt:

How can I increase shop sales?

AI will give **general advice**.



## **Improved Multi-Step Prompt**

You are a business consultant.

Help a small shop owner increase sales.

Follow this process:

Step 1: Identify possible reasons for low sales

Step 2: Suggest improvement strategies

Step 3: Estimate possible results

Step 4: Give final recommendation

Output format:

Problem Analysis:

Improvement Strategies:

Expected Results:

Final Recommendation:

## **Why This Prompt Works**

It guides AI to:

1. analyze problem
2. suggest solutions
3. evaluate results
4. give final decision

This produces **better strategic answers**.

## Real-Life Example

Prompt:

You are a career advisor.

Help a student choose the best career.

Follow this process:

1. Identify student interests
2. Compare career options
3. Evaluate future demand
4. Recommend the best option

Output format:

Student Interests:

Career Options:

Evaluation:

Final Recommendation:

## Where This Technique Is Used

- business strategy
- consulting tools
- AI decision systems

- career guidance systems

## Your Practice Task

Improve this prompt using **multi-step problem solving**.

Bad prompt:

How can a student improve study habits?

Try to include:

- role
- step-by-step process
- structured output

## Exercise 27: Prompt for Creative Thinking

### Core Idea

Prompt engineering can help AI generate **creative ideas**.

This is useful for:

- business ideas
- startup concepts
- marketing campaigns
- product innovation
- content ideas

But prompts must encourage **creative thinking**.

### Example 1: Weak Prompt

Give business ideas.

Problem:

- Too vague
- No industry

- No audience

## **Improved Creative Prompt**

You are a startup innovation expert.

Generate 5 creative business ideas related to Artificial Intelligence.

Constraints:

- Ideas should be suitable for beginners
- Each idea should require low investment

Output format:

Idea 1:

Problem solved:

How it works:

Idea 2:

Problem solved:

How it works:

## **Why This Prompt Works**

It clearly defines:

- role → innovation expert
- industry → AI

- constraint → low investment
- structured idea format

This leads to **better and more practical ideas**.

## Real-Life Example

Prompt:

You are a marketing strategist.

Generate 5 creative marketing ideas to promote a new Python course.

Constraints:

- Suitable for college students
- Budget under ₹10,000

Output format:

Idea

Execution method

Expected impact

## Where This Prompt Type Is Used

- startup idea generation
- product brainstorming
- marketing strategy

- innovation workshops

## **Your Practice Task**

Improve this prompt:

Give creative ideas for a business.

Try to include:

- role
- industry
- constraints
- output format

## Exercise 28: Prompt for Comparison

### Core Idea

Comparison prompts help AI **analyze differences between options**.

This is useful for:

- product comparison
- technology comparison
- career choices
- business decisions

Prompt must clearly define:

- items to compare
- comparison criteria
- output format

### Example 1: Weak Prompt



Compare Python and Java.

Problem:

- No criteria
- No structure

### **Improved Prompt**

You are a software engineering mentor.

Compare Python and Java programming languages.

Comparison criteria:

- learning difficulty
- job demand
- performance
- common use cases

Output format:

Comparison Table:

| Criteria | Python | Java |
|----------|--------|------|
|----------|--------|------|

Then provide:

1. Advantages of Python
2. Advantages of Java
3. Final recommendation for beginners

## Why This Prompt Works

It clearly defines:

- role → software mentor
- comparison items → Python vs Java
- criteria → learning, jobs, performance
- format → table + explanation

## Real-Life Example

Prompt:

You are a career advisor.

Compare Data Science and Software Development careers.

Criteria:

- salary potential
- learning difficulty
- job demand
- career growth

Output format:

Comparison table

Pros and cons

Final suggestion

### **Where This Prompt Type Is Used**

- product reviews
- technology decisions
- career guidance
- business planning

### **Your Practice Task**

Improve this prompt:

Compare two programming languages.

Try to include:

- role
- languages to compare
- comparison criteria
- output format

## Exercise 29: Prompt for Error Detection

### Core Idea

AI can help **detect mistakes** in:

- code
- grammar
- logic
- calculations
- written content

Prompt must clearly specify:

- what type of errors to find
- what correction is required
- output format

### Example 1: Weak Prompt

Check this code.

Problem:

- Not clear what to check
- No format for correction

## Improved Prompt

You are a Python code reviewer.

Analyze the following Python code.

Tasks:

1. Identify errors in the code
2. Explain why the error occurs
3. Provide the corrected code

Output format:

Error Found:

Explanation:

Corrected Code:

## Why This Prompt Works

It clearly defines:

- role → code reviewer
- task → find errors
- correction required

- structured output

## **Real-Life Example**

Prompt:

You are a grammar expert.

Check the following paragraph.

Tasks:

- Identify grammar mistakes
- Correct the sentence

Output format:

Incorrect sentence:

Correct sentence:

Explanation:

## **Where This Prompt Type Is Used**

- coding assistants
- grammar correction tools
- document editing systems
- debugging tools

## **Your Practice Task**

Improve this prompt:

Find mistakes in this text.

Try to include:

- role
- type of mistakes
- output format

## Exercise 30: Prompt for Workflow Automation

### Core Idea

Workflow automation means **AI completes multiple tasks in sequence**.

Instead of one task, AI performs a **small workflow**.

Example workflow:

Input → Process → Output

Prompt engineers design prompts to **automate repetitive tasks**.

### Example: Study Content Workflow

Weak prompt:

Explain machine learning.

Problem:

- Only one output



- No workflow

## Improved Workflow Prompt

You are an AI study assistant.

Process the following topic using this workflow.

Topic: Machine Learning

Step 1: Explain the topic in simple English

Step 2: Create a 5-point summary

Step 3: Generate 3 multiple-choice questions with answers

Output format:

Explanation:

Summary:

1)

2)

3)

4)

5)

MCQs:

Question 1:

Options:

Answer:

## Why This Prompt Works

It defines a **mini automation system**:

Explain → Summarize → Generate questions

This technique is widely used in:

- learning platforms
- content generators
- productivity tools

## Real-Life Example

Prompt:

You are an AI resume assistant.

Process the resume using this workflow.

Step 1: Extract candidate information

Step 2: Identify strengths and weaknesses

Step 3: Generate 5 interview questions

Output format:

Candidate Details:

Strengths:

Weaknesses:

Interview Questions:

## Where This Is Used

- AI productivity tools
- study assistants
- HR automation
- content automation

## Your Practice Task

Improve this prompt:

Help students learn Python.

Try to create a **workflow prompt** with steps like:

- explanation
- summary
- questions

## Exercise 31: Prompt for Knowledge Expansion

### Core Idea

Knowledge expansion prompts help AI **teach a topic deeply**.

Instead of giving only a short explanation, AI will:

1. explain the concept
2. expand related ideas
3. connect with practical applications

This helps students **understand the full picture**.

### Example 1: Weak Prompt

Explain machine learning.

Problem:

- Only basic explanation
- No deeper learning

## Improved Prompt

You are an AI instructor.

Teach the topic **Machine Learning** step by step.

Tasks:

1. Explain the basic concept
2. Explain related concepts
3. Give practical applications
4. Suggest topics to learn next

Output format:

Basic Explanation:

Related Concepts:

Real-Life Applications:

Next Topics to Learn:

## Why This Prompt Works

It expands knowledge in **layers**:

Concept → Related ideas → Applications → Next learning path  
This is very useful for **education systems**.

## Real-Life Example

Prompt:

You are a programming mentor.

Teach the topic **Python functions**.

Tasks:

1. Explain the concept
2. Give simple examples
3. Show real-world uses
4. Suggest advanced topics

Output format:

Concept:

Examples:

Real-world uses:

Next topics:

## Where This Prompt Type Is Used

- AI tutors

- education platforms
- research assistants
- learning systems

## **Your Practice Task**

Improve this prompt:

Teach me about data science.

Try to include:

- role
- layered explanation
- output format

## Exercise 32: Prompt for Idea Expansion

### Core Idea

Sometimes we already have **one idea**, but we want to **expand it into many ideas**.

Prompt engineering can help AI:

- generate variations
- expand possibilities
- explore new angles

This technique is used for **brainstorming and innovation**.

### Example 1: Weak Prompt

Give startup ideas.

Problem:

- Too broad



- No direction

## **Improved Prompt**

You are a startup innovation expert.

Expand the following idea into multiple related business ideas.

Base idea: AI-based education platform

Tasks:

1. Generate 5 related startup ideas
2. Explain how each idea works
3. Mention the target audience

Output format:

Idea 1:

Description:

Target users:

Idea 2:

Description:

Target users:

## **Why This Prompt Works**

It takes **one base idea** and expands it into **multiple possibilities**.

This technique helps in:

- startup planning
- product development
- marketing campaigns

## Real-Life Example

Prompt:

You are a product strategist.

Expand the idea "**online coding courses**" into 5 new product ideas.

Output format:

Idea

How it works

Who will use it

## Where This Prompt Type Is Used

- startup brainstorming
- product innovation
- marketing strategy
- research exploration

## **Your Practice Task**

Improve this prompt:

Expand this business idea.

Try to include:

- role
- base idea
- number of ideas
- output format

## Exercise 33: Prompt for Scenario Simulation

### Core Idea

Scenario simulation means **creating a realistic situation and asking AI to simulate it.**

AI can act like:

- an interviewer
- a customer
- a manager
- a teacher
- a business partner

This helps users **practice real-world situations.**

### Example 1: Weak Prompt

Ask me interview questions.

Problem:

- No role
- No job position
- No interaction

### Improved Prompt

You are a technical interviewer.

Conduct a mock interview for a Python developer.

Process:

1. Ask one interview question at a time
2. Wait for my answer
3. Evaluate my answer
4. Suggest improvements

Output format:

Question:

Evaluation:

Improvement suggestion:

### Why This Prompt Works

It creates a **real interview simulation**.

This is very useful for:

- interview practice
- sales training
- negotiation training
- leadership training

## **Real-Life Example**

Prompt:

You are a customer support trainer.

Simulate a conversation with an angry customer.

Process:

1. Customer explains the complaint
2. I respond as the support agent
3. Evaluate my response

Output format:

Customer message:

My response:

Evaluation:

## **Where This Technique Is Used**

- interview preparation tools
- training simulators

- leadership coaching
- sales training systems

## **Your Practice Task**

Improve this prompt:

Help me practice a job interview.

Try to include:

- role
- scenario
- interaction steps
- output format

## Exercise 34: Prompt for Structured Learning

### Core Idea

Structured learning prompts help AI **teach topics in an organized way.**

Instead of random explanation, AI teaches like a **teacher in a classroom.**

Structure may include:

- introduction
- key concepts
- examples
- summary
- practice questions



## Example 1: Weak Prompt

Teach me Python.

Problem:

- No structure
- Random explanation

## Improved Prompt

You are a programming teacher.

Teach the topic **Python variables**.

Follow this teaching structure:

1. Simple explanation
2. 3 key concepts
3. 1 real-life example
4. Quick summary
5. 2 practice questions

Output format:

Explanation:

Key Concepts:

Example:

Summary:

Practice Questions:

## Why This Prompt Works

It turns AI into a **structured learning system**.

This method is used in:

- AI tutors
- learning platforms
- study assistants
- course creation systems

## Real-Life Example

Prompt:

You are a data science instructor.

Teach the topic **linear regression**.

Teaching structure:

1. Concept explanation
2. Key formula
3. Example use case

4. Summary

5. Practice question

### **Where This Prompt Type Is Used**

- online courses
- AI teaching assistants
- exam preparation tools
- study note generators

### **Your Practice Task**

Improve this prompt:

Teach me machine learning.

Try to include:

- role
- structured learning steps
- output format

## Exercise 35: Prompt for Expert Advice

### Core Idea

Sometimes we want AI to give **expert-level advice**.

Prompt engineers simulate an **experienced professional**.

AI can act like:

- business consultant
- career coach
- marketing strategist
- financial advisor
- startup mentor

This helps users get **practical guidance**.

### Example 1: Weak Prompt

Give business advice.

Problem:

- No industry
- No situation
- No expertise level

### **Improved Prompt**

You are a startup mentor with 20 years of experience.

Advise a beginner entrepreneur who wants to start a small online business.

Tasks:

1. Suggest 5 important steps
2. Explain common mistakes
3. Provide one real-life example

Output format:

Steps to follow:

Common mistakes:

Example:

### **Why This Prompt Works**

It clearly defines:

- role → experienced mentor
- situation → beginner entrepreneur

- tasks → steps, mistakes, example
- structured output

This produces **practical advice instead of generic text**.

## Real-Life Example

Prompt:

You are a career coach.

Advise a college student who wants to become a data scientist.

Tasks:

1. Skills to learn
2. Learning roadmap
3. Career tips

Output format:

Skills:

Roadmap:

Career tips:

## Where This Prompt Type Is Used

- consulting tools
- career guidance systems
- business strategy tools

- mentoring platforms

## **Your Practice Task**

Improve this prompt:

Give advice for learning programming.

Try to include:

- expert role
- target audience
- advice structure
- output format

## Exercise 36: Prompt for Knowledge Testing

### Core Idea

Knowledge testing prompts are used to **check understanding of a topic**.

AI can generate:

- quizzes
- MCQs
- short-answer questions
- problem-solving tasks

This is widely used in:

- online learning platforms
- exam preparation apps
- AI tutors



### Example 1: Weak Prompt

Ask questions about Python.

Problem:

- No difficulty level
- No number of questions
- No format

### Improved Prompt

You are an exam question creator.

Generate questions to test knowledge of **Python programming basics**.

Constraints:

- Create 5 multiple-choice questions
- Difficulty level: beginner

Output format:

Question:

A) Option

B) Option

C) Option

D) Option

Answer:

## Why This Prompt Works

It clearly defines:

- role → exam creator
- topic → Python basics
- question type → MCQs
- difficulty → beginner
- format → structured output

## Real-Life Example

Prompt:

You are a data science instructor.

Create 5 quiz questions on **machine learning basics**.

Constraints:

- beginner level
- include correct answers

Output format:

Question:

Options:

Answer:

### **Where This Prompt Type Is Used**

- AI tutors
- exam preparation systems
- online quizzes
- interview practice tools

### **Your Practice Task**

Improve this prompt:

Test my knowledge of AI.

Try to include:

- role
- number of questions
- difficulty level
- output format

## Exercise 37: Prompt for Feedback Generation

### Core Idea

Feedback prompts help AI **evaluate performance and give suggestions**.

AI can provide feedback for:

- interview answers
- essays
- presentations
- coding assignments
- student work

Prompt must clearly specify:

- what to evaluate
- evaluation criteria
- feedback format

### **Example 1: Weak Prompt**

Give feedback on this answer.

Problem:

- No criteria
- No evaluation structure

### **Improved Prompt**

You are an interview evaluator.

Review the following candidate answer.

Tasks:

1. Evaluate clarity
2. Evaluate correctness
3. Suggest improvements

Output format:

Strengths:

Weaknesses:

Suggestions for improvement:

## Why This Prompt Works

It clearly defines:

- role → evaluator
- criteria → clarity, correctness
- structured feedback

This produces **useful improvement advice**.

## Real-Life Example

Prompt:

You are a programming mentor.

Review the following Python code written by a student.

Tasks:

- identify mistakes
- suggest improvements

Output format:

Correct parts:

Issues found:

Improvement suggestions:

### **Where This Prompt Type Is Used**

- AI coding mentors
- interview practice tools
- education platforms
- coaching systems

### **Your Practice Task**

Improve this prompt:

Review my project.

Try to include:

- role
- evaluation criteria
- output format

## Exercise 38: Prompt for Personalization

### Core Idea

Personalization prompts help AI **customize responses for a specific person**.

Instead of giving **general advice**, AI adapts the answer based on:

- user background
- skill level
- goals
- situation

This makes AI responses **more useful and practical**.

### Example 1: Weak Prompt



Give career advice.

Problem:

- No user information
- Advice becomes generic

### **Improved Prompt**

You are a career advisor.

Provide career guidance for a student with the following profile:

Background:

- B.Tech Computer Science student
- Beginner in programming
- Interested in Artificial Intelligence

Tasks:

1. Suggest skills to learn
2. Suggest learning resources
3. Suggest career paths

Output format:

Skills to learn:

Learning resources:

Career paths:

## Why This Prompt Works

It includes **personal context**, so AI gives **custom advice**.

## Real-Life Example

Prompt:

You are a fitness coach.

Create a fitness plan for a person with the following profile:

Age: 25

Goal: weight loss

Exercise level: beginner

Output format:

Workout plan:

Diet tips:

Weekly goal:

## Where This Prompt Type Is Used

- AI personal assistants
- education platforms
- career guidance systems

- fitness coaching apps

### **Your Practice Task**

Improve this prompt:

Suggest a learning plan.

Try to include:

- role
- user profile
- tasks
- output format

## Exercise 39: Prompt for Knowledge Organization

### Core Idea

Sometimes information is **scattered and messy**.

Prompt engineering can help AI **organize knowledge clearly**.

AI can organize information into:

- categories
- tables
- structured lists
- learning frameworks

This is very useful for:

- study materials
- documentation
- research notes

**Example 1: Weak Prompt**

Organize this information.

Problem:

- No structure
- No format defined

**Improved Prompt**

You are a study organizer.

Organize the following information about Artificial Intelligence.

Tasks:

1. Group related concepts together
2. Arrange topics in logical order

Output format:

Category 1: Basics

Topics:

Category 2: Core Techniques

Topics:

Category 3: Applications

Topics:

## Why This Prompt Works

It converts **random information** → **structured knowledge**.

This helps students **learn faster**.

## Real-Life Example

Prompt:

You are a technical documentation expert.

Organize the following programming topics.

Tasks:

- group similar topics
- create learning sequence

Output format:

Beginner topics:

Intermediate topics:

Advanced topics:

## Where This Prompt Type Is Used

- learning platforms
- documentation tools

- research assistants
- knowledge management systems

### **Your Practice Task**

Improve this prompt:

Organize these study topics.

Try to include:

- role
- organization criteria
- output format

## Exercise 40: Prompt for Workflow Planning

### Core Idea

Workflow planning prompts help AI **create a step-by-step plan to complete a task.**

This is useful for:

- project planning
- study planning
- business execution
- product development

Prompt engineers ask AI to **break big goals into clear steps.**

### Example 1: Weak Prompt

How do I build a website?



Problem:

- Too broad
- No structure
- No step-by-step plan

### **Improved Prompt**

You are a project planner.

Create a step-by-step workflow to build a simple website.

Tasks:

1. Identify the main stages
2. List steps in each stage
3. Suggest tools required

Output format:

Stage 1: Planning

Steps:

Stage 2: Development

Steps:

Stage 3: Testing

Steps:

Stage 4: Deployment

Steps:

## Why This Prompt Works

It converts a **big task into clear stages and steps**.

This improves:

- clarity
- execution
- productivity

## Real-Life Example

Prompt:

You are a study planner.

Create a 30-day learning plan for Python programming.

Tasks:

- divide topics week-wise
- suggest daily practice tasks

Output format:

Week 1:

Topics:

Practice tasks:

Week 2:

Topics:

Practice tasks:

### **Where This Prompt Type Is Used**

- project management tools
- productivity assistants
- study planning systems
- business planning tools

### **Your Practice Task**

Improve this prompt:

Plan a project.

Try to include:

- role
- project type
- workflow stages
- output format

## Exercise 41: Prompt for Strategy Development

### Core Idea

Strategy prompts help AI **create a plan to achieve a goal**.

Unlike simple advice, strategy includes:

- analysis
- planning
- execution steps

This type of prompt is used in:

- business strategy
- marketing planning
- career planning
- product growth

### **Example 1: Weak Prompt**

How can I grow my business?

Problem:

- Too broad
- No industry
- No structure

### **Improved Prompt**

You are a business strategist.

Help develop a growth strategy for a small online education platform.

Tasks:

1. Analyze current challenges
2. Suggest growth strategies
3. Identify marketing methods
4. Recommend next steps

Output format:

Current challenges:

Growth strategies:

Marketing methods:

Next steps:

## Why This Prompt Works

It guides AI to:

- analyze the situation
- propose strategy
- give actionable steps

This produces **practical strategic advice**.

## Real-Life Example

Prompt:

You are a digital marketing strategist.

Create a marketing strategy for a new Python programming course.

Tasks:

- identify target audience
- suggest marketing channels
- recommend campaign ideas

Output format:

Target audience:

Marketing channels:

Campaign ideas:

### **Where This Prompt Type Is Used**

- business consulting tools
- marketing automation
- startup planning
- product strategy

### **Your Practice Task**

Improve this prompt:

Create a marketing strategy.

Try to include:

- role
- product/service
- tasks
- output format

## Exercise 42: Prompt for Problem Diagnosis

### Core Idea

Problem diagnosis prompts help AI **identify the root cause of a problem**.

Instead of directly giving solutions, AI first:

1. understands the problem
2. analyzes possible causes
3. suggests solutions

This is widely used in:

- troubleshooting systems
- technical support



- consulting tools
- debugging tools

### **Example 1: Weak Prompt**

My website is slow. What should I do?

Problem:

- AI may give random suggestions
- No structured analysis

### **Improved Prompt**

You are a website performance expert.

Diagnose the problem of a slow website.

Tasks:

1. Identify possible causes
2. Explain how each cause affects performance
3. Suggest solutions

Output format:

Possible causes:

Explanation:

Recommended solutions:

## Why This Prompt Works

It forces AI to **analyze the problem before giving solutions**.

This improves:

- accuracy
- clarity
- usefulness

## Real-Life Example

Prompt:

You are a technical support engineer.

Diagnose why a Python program is not running correctly.

Tasks:

- identify possible errors

- explain the cause
- suggest fixes

Output format:

Problem cause:

Explanation:

Suggested fix:

### **Where This Prompt Type Is Used**

- technical troubleshooting tools
- IT support systems
- debugging assistants
- consulting platforms

### **Your Practice Task**

Improve this prompt:

My laptop is slow. Help me.

Try to include:

- role
- diagnostic steps
- output format

## Exercise 43: Prompt for Learning Roadmaps

### Core Idea

Learning roadmap prompts help AI **design a structured path to learn a skill**.

Instead of random topics, AI creates:

- step-by-step learning plan
- topic sequence
- practice tasks

This is very useful for:

- career planning
- skill development

- online courses
- self-learning

### **Example 1: Weak Prompt**

How can I learn data science?

Problem:

- AI gives general advice
- No learning sequence

### **Improved Prompt**

You are a career mentor.

Create a learning roadmap to become a data scientist.

Tasks:

1. Identify beginner topics
2. Identify intermediate topics
3. Identify advanced topics
4. Suggest practical projects

Output format:

Beginner stage:

Topics:

Intermediate stage:

Topics:

Advanced stage:

Topics:

Projects:

## Why This Prompt Works

It creates a **clear learning path instead of random advice**.

This helps learners **progress step by step**.

## Real-Life Example

Prompt:

You are a programming mentor.

Create a 3-month roadmap to learn Python programming.

Tasks:

- divide topics week-wise
- suggest practice exercises

Output format:

Month 1:

Topics:

Month 2:

Topics:

Month 3:

Topics:

Practice exercises:

### **Where This Prompt Type Is Used**

- career guidance tools
- learning platforms
- skill development programs
- education apps

### **Your Practice Task**

Improve this prompt:

Create a learning plan for AI.

Try to include:

- role
- learning stages

- tasks
- output format

## Exercise 44: Prompt for Decision Trees

### Core Idea

Decision tree prompts help AI **guide users through decisions step-by-step**.

Instead of giving one answer, AI creates a **decision flow**.

This is useful for:

- career guidance
- troubleshooting systems
- medical triage tools
- customer support systems



### **Example 1: Weak Prompt**

Help me choose a programming language.

Problem:

- AI gives one general answer
- No guided decision process

### **Improved Prompt**

You are a programming mentor.

Create a decision tree to help beginners choose a programming language.

Decision factors:

- career goal
- interest in web development
- interest in data science

Output format:

Step 1: Ask the user about their goal

If goal = Web Development → Suggest JavaScript

Step 2: Ask about interest in Data Science

If yes → Suggest Python

Step 3: Ask about mobile development  
If yes → Suggest Kotlin or Swift

### Why This Prompt Works

It creates a **guided decision process** instead of a simple answer.  
This helps users **choose based on their situation**.

### Real-Life Example

Prompt:

You are a career advisor.

Create a decision tree to help students choose between:

- Software Development
- Data Science
- Cybersecurity

Output format:

Step 1: Ask about interest

Step 2: Suggest options based on answers

Step 3: Provide final recommendation

### Where This Prompt Type Is Used

- AI career advisors
- technical troubleshooting bots
- decision support systems
- customer support automation

### **Your Practice Task**

Improve this prompt:

Help me choose a career.

Try to include:

- role
- decision factors
- step-by-step decision flow
- output format

## Exercise 45: Prompt for Knowledge Simplification

### Core Idea

Some topics are **complex and difficult**.

Prompt engineers can ask AI to **simplify the topic**.

AI can explain using:

- simple language
- analogies
- real-life examples

This helps beginners **understand complex ideas easily**.

## Example 1: Weak Prompt

Explain neural networks.

Problem:

- Explanation may be very technical
- Beginners may not understand

## Improved Prompt

You are a friendly teacher.

Explain **neural networks** to a beginner who has no background in AI.

Constraints:

- Use very simple English
- Use a real-life analogy

Output format:

Simple explanation:

Real-life analogy:

Key idea summary:

## Why This Prompt Works

It ensures the explanation becomes:

- beginner friendly
- easy to understand
- practical

## Real-Life Example

Prompt:

You are a science teacher.

Explain **blockchain technology** to a school student.

Constraints:

- Use simple words
- Give a daily-life example

Output format:

Explanation:

Example:

Summary:

## Where This Prompt Type Is Used

- education platforms
- AI tutors
- knowledge assistants
- beginner learning tools

### **Your Practice Task**

Improve this prompt:

Explain artificial intelligence.

Try to include:

- role
- beginner audience
- simplification instructions
- output format

## Exercise 46: Prompt for Idea Evaluation

### Core Idea

Sometimes we have an **idea**, but we don't know if it is good or not.

Prompt engineering can help AI **evaluate an idea logically**.

AI can analyze:

- feasibility
- benefits
- risks
- improvements



This is useful in:

- startup planning
- product development
- innovation evaluation

### **Example 1: Weak Prompt**

Is this business idea good?

Problem:

- No evaluation criteria
- AI gives vague answer

### **Improved Prompt**

You are a startup advisor.

Evaluate the following business idea.

Idea:

An AI-powered study assistant for college students.

Evaluation criteria:

1. Market demand
2. Potential benefits
3. Possible risks
4. Suggestions for improvement

Output format:

Idea summary:

Benefits:

Risks:

Improvement suggestions:

## Why This Prompt Works

It forces AI to **analyze the idea systematically**.

This produces **clear and actionable insights**.

## Real-Life Example

Prompt:

You are a product strategist.

Evaluate the idea of launching an online Python coding bootcamp.

Evaluation criteria:

- target market

- competition
- revenue potential
- improvement suggestions

Output format:

Market analysis:

Competition:

Revenue potential:

Suggestions:

### **Where This Prompt Type Is Used**

- startup incubators
- product design teams
- innovation labs
- business consulting

### **Your Practice Task**

Improve this prompt:

Evaluate this startup idea.

Try to include:

- role
- evaluation criteria

- output format

## Exercise 47: Prompt for Role-Playing Conversations

### Core Idea

Role-playing prompts make AI **act like a specific person in a conversation.**

This helps users practice **real situations.**

AI can act as:

- customer
- interviewer
- manager
- teacher

- client

This is useful for:

- interview practice
- sales training
- negotiation training
- communication improvement

### **Example 1: Weak Prompt**

Talk to me like a customer.

Problem:

- No scenario
- No interaction steps

### **Improved Prompt**

You are a customer interested in buying a laptop.

We will role-play a sales conversation.

Process:

1. You ask questions as a customer
2. I respond as the sales representative

### 3. You evaluate my response

Output format:

Customer question:

My response:

Evaluation:

### Why This Prompt Works

It creates a **real conversation simulation**.

This helps users **practice communication skills**.

### Real-Life Example

Prompt:

You are an HR interviewer.

Conduct a job interview for a software developer role.

Process:

1. Ask one question at a time
2. Wait for my answer
3. Evaluate the answer

Output format:

Question:

Evaluation:

Suggestion:

### **Where This Prompt Type Is Used**

- interview simulators
- sales training tools
- communication coaching
- negotiation training systems

### **Your Practice Task**

Improve this prompt:

Practice a sales conversation with me.

Try to include:

- role
- scenario
- interaction steps
- output format

## Exercise 48: Prompt for Task Decomposition

### Core Idea

Sometimes a task is **very complex**.

Instead of solving it directly, we **break it into smaller tasks**.

This is called **task decomposition**.

Prompt engineers guide AI to:

1. divide the task
2. solve each part



### 3. combine the result

#### **Example 1: Weak Prompt**

Build a mobile app.

Problem:

- Too big task
- No breakdown

#### **Improved Prompt**

You are a software project planner.

Break down the task of building a mobile app.

Tasks:

1. Identify major components
2. Break each component into smaller tasks
3. Arrange tasks in logical order

Output format:

Component 1:

Subtasks:

Component 2:

Subtasks:

Component 3:

Subtasks:

## Why This Prompt Works

It converts **big problems** → **manageable tasks**.

This helps in:

- project management
- product development
- learning complex skills

## Real-Life Example

Prompt:

You are a study planner.

Break down the task of learning **data science**.

Tasks:

- identify major topics
- divide them into smaller learning tasks

Output format:

Topic:

Subtopics:

Practice tasks:

### **Where This Prompt Type Is Used**

- AI planning systems
- project management tools
- learning systems
- complex problem solving

### **Your Practice Task**

Improve this prompt:

Help me build a startup.

Try to include:

- role
- task decomposition steps
- output format

## Exercise 49: Prompt for Pattern Recognition

### Core Idea

Pattern recognition prompts help AI **identify trends or repeated patterns in data**.

AI can analyze:

- sales trends
- user behavior

- learning performance
- customer feedback

This is useful for:

- business analytics
- marketing insights
- data science tasks

### **Example 1: Weak Prompt**

Analyze this data.

Problem:

- No task defined
- AI may give vague analysis

### **Improved Prompt**

You are a data analyst.

Analyze the following monthly sales data and identify patterns.

Data:

January: ₹50,000

February: ₹60,000

March: ₹75,000

April: ₹80,000

May: ₹78,000

Tasks:

1. Identify sales trend
2. Detect any unusual pattern
3. Suggest possible reasons

Output format:

Trend observed:

Unusual patterns:

Possible reasons:

## Why This Prompt Works

It clearly defines:

- role → data analyst
- data → provided numbers
- analysis tasks → trend + anomaly detection

## Real-Life Example

Prompt:

You are a learning analyst.

Analyze student exam scores.

Tasks:

- identify performance trends
- detect weak areas

Output format:

Trend:

Weak topics:

Improvement suggestions:

### **Where This Prompt Type Is Used**

- business analytics
- AI data analysis tools
- education analytics
- marketing insights

### **Your Practice Task**

Improve this prompt:

Find patterns in this data.

Try to include:

- role
- data type
- analysis tasks
- output format

## Exercise 50: Prompt for Recommendation Systems

### Core Idea

Recommendation prompts help AI **suggest the best options based on user needs.**

AI can recommend:

- books
- courses
- products
- learning resources



- career paths

Prompt must include:

- user profile
- preferences
- recommendation criteria

### **Example 1: Weak Prompt**

Recommend a programming course.

Problem:

- No user background
- No learning goal

### **Improved Prompt**

You are a career mentor.

Recommend programming courses for the following learner profile.

User profile:

- College student
- Beginner in programming
- Interested in Artificial Intelligence

Tasks:

1. Suggest 3 suitable courses
2. Explain why each course is useful
3. Suggest learning order

Output format:

Course 1:

Reason:

Course 2:

Reason:

Course 3:

Reason:

Learning order:

## **Why This Prompt Works**

It includes:

- user context
- recommendation logic
- structured output

This produces **more relevant suggestions**.

## Real-Life Example

Prompt:

You are a book recommendation expert.

Suggest books for someone who wants to learn about personal finance.

Constraints:

- beginner friendly

Output format:

Book name:

Why it is useful:

## Where This Prompt Type Is Used

- AI assistants
- e-commerce recommendation systems
- learning platforms
- streaming platforms

## Your Practice Task

Improve this prompt:

Recommend books for me.

Try to include:

- role
- user interests
- recommendation criteria
- output format

## Exercise 51: Prompt for Multi-Perspective Analysis

### Core Idea

Sometimes a problem must be analyzed from **multiple perspectives**.

AI can evaluate an issue from different viewpoints like:

- customer
- business owner
- investor
- employee
- technical expert

This helps make **better decisions**.

### Example 1: Weak Prompt

Is this product good?

Problem:

- Only one viewpoint
- Limited analysis

### Improved Prompt

You are a business analyst.

Analyze the idea of launching an **AI-based study app** from multiple perspectives.

Perspectives to analyze:

1. Customer perspective
2. Business owner perspective
3. Investor perspective
4. Technical feasibility

Output format:

Customer perspective:

Business perspective:

Investor perspective:

Technical feasibility:

## Why This Prompt Works

It forces AI to think from **different angles**, producing **deeper insights**.

## Real-Life Example

Prompt:

You are a strategy consultant.

Analyze the idea of opening an online coding school.

Perspectives:

- student perspective
- instructor perspective
- business perspective

Output format:

Student perspective:

Instructor perspective:

Business perspective:

## **Where This Prompt Type Is Used**

- strategy consulting
- product planning
- investment analysis
- business decision making

## **Your Practice Task**

Improve this prompt:

Analyze this business idea.

Try to include:

- role
- perspectives to analyze
- output format

## Exercise 52: Prompt for Ethical Evaluation

### Core Idea

Some ideas or technologies must be checked from an **ethical perspective**.

AI can help evaluate:

- social impact
- ethical risks
- responsible use

This is important in:



- Artificial Intelligence
- healthcare
- finance
- public systems

Prompt engineers guide AI to **analyze ethical consequences**.

### **Example 1: Weak Prompt**

Is this technology good?

Problem:

- No ethical evaluation
- Only general opinion

### **Improved Prompt**

You are an AI ethics expert.

Evaluate the ethical implications of using facial recognition technology.

Tasks:

1. Identify potential benefits
2. Identify ethical risks
3. Suggest responsible usage guidelines

Output format:

Benefits:

Ethical risks:

Responsible usage guidelines:

## Why This Prompt Works

It ensures technology is analyzed from **ethical and social perspectives**, not just technical benefits.

## Real-Life Example

Prompt:

You are a technology ethics advisor.

Evaluate the use of AI in hiring systems.

Tasks:

- identify potential bias risks
- suggest ethical safeguards

Output format:

Benefits:

Risks:

Safeguards:

### **Where This Prompt Type Is Used**

- AI governance
- policy development
- responsible AI systems
- technology regulation

### **Your Practice Task**

Improve this prompt:

Is AI good or bad?

Try to include:

- role
- ethical evaluation criteria
- output format

## Exercise 53: Prompt for Knowledge Integration

### Core Idea

Sometimes we learn **multiple topics separately**, but we need to **connect them together**.

Knowledge integration prompts help AI:

- combine different ideas
- explain relationships
- show how topics work together

This is useful in:

- education
- research
- interdisciplinary learning

### **Example 1: Weak Prompt**

Explain AI and data science.

Problem:

- Topics explained separately
- No connection shown

### **Improved Prompt**

You are a technology instructor.

Explain how the following topics are connected:

- Artificial Intelligence
- Machine Learning
- Data Science

Tasks:

1. Explain each topic briefly
2. Explain how they are related
3. Give a real-world example

Output format:

Topic explanations:

Relationship between topics:

Real-world example:

### Why This Prompt Works

It helps learners **understand the bigger picture** instead of isolated knowledge.

### Real-Life Example

Prompt:

You are a programming mentor.

Explain how **Python, Machine Learning, and Data Analysis** work together.

Output format:

Explanation of each concept:

How they connect:

Example project:

### Where This Prompt Type Is Used

- AI education platforms

- research analysis
- interdisciplinary learning
- technical training

### **Your Practice Task**

Improve this prompt:

Explain how technology topics are related.

Try to include:

- role
- specific topics
- tasks
- output format

## Exercise 54: Prompt for Future Predictions

### Core Idea

Future prediction prompts ask AI to **analyze current trends and suggest possible future developments**.

AI cannot predict the future perfectly, but it can:

- analyze trends
- identify patterns
- suggest possible scenarios

This is useful in:



- business strategy
- technology forecasting
- market research
- career planning

### **Example 1: Weak Prompt**

What is the future of AI?

Problem:

- Too vague
- No structured analysis

### **Improved Prompt**

You are a technology analyst.

Analyze the future of Artificial Intelligence in the next 10 years.

Tasks:

1. Identify current trends in AI
2. Predict possible developments
3. Suggest potential challenges

Output format:

Current trends:

Future developments:

Possible challenges:

## **Why This Prompt Works**

It guides AI to:

- analyze present trends
- project future possibilities
- evaluate risks

## **Real-Life Example**

Prompt:

You are a business strategist.

Predict the future of online education in the next 5 years.

Tasks:

- identify current trends
- suggest future opportunities

Output format:

Current trends:

Future opportunities:

Challenges:

## **Where This Prompt Type Is Used**

- technology forecasting
- business strategy
- research analysis
- career guidance

### **Your Practice Task**

Improve this prompt:

Predict the future of technology.

Try to include:

- role
- prediction timeframe
- analysis tasks
- output format

## Exercise 55: Prompt for Scenario Planning

### Core Idea

Scenario planning means **thinking about different possible future situations**.

Instead of predicting only one future, AI analyzes:

- best-case scenario
- worst-case scenario
- most likely scenario

This helps in:

- business strategy
- risk management
- planning under uncertainty

### **Example 1: Weak Prompt**

What will happen to my business?

Problem:

- No structured analysis
- Only general advice

### **Improved Prompt**

You are a business strategist.

Analyze possible future scenarios for an online education business.

Tasks:

1. Describe the best-case scenario
2. Describe the worst-case scenario
3. Describe the most likely scenario

Output format:

Best-case scenario:

Worst-case scenario:

Most likely scenario:

### **Why This Prompt Works**

It helps decision makers **prepare for different possibilities**.

### **Real-Life Example**

Prompt:

You are a technology strategist.

Analyze possible scenarios for the future of AI-powered automation.

Output format:

Best-case scenario:

Worst-case scenario:

Most likely scenario:

### **Where This Prompt Type Is Used**

- strategic planning
- business forecasting
- risk analysis
- policy planning

## Your Practice Task

Improve this prompt:

What will happen in the future?

Try to include:

- role
- topic
- scenario types
- output format

## Exercise 56: Prompt for Decision Justification

### Core Idea

Sometimes AI gives a recommendation, but we also want to know **why that decision was made**.

Decision justification prompts make AI:

1. give a recommendation
2. explain the reasoning
3. provide supporting arguments

This improves **trust and clarity**.

### **Example 1: Weak Prompt**

Which programming language should I learn?

Problem:

- AI gives one suggestion
- No reasoning explained

### **Improved Prompt**

You are a programming mentor.

Recommend the best programming language for a beginner who wants to work in Artificial Intelligence.

Tasks:

1. Suggest the best language
2. Explain why this language is suitable
3. Mention advantages of this choice

Output format:



Recommended language:

Reason for recommendation:

Advantages:

## Why This Prompt Works

It ensures the answer includes **justification**, not just a suggestion.

## Real-Life Example

Prompt:

You are a career advisor.

Recommend the best career path for someone interested in data analysis.

Tasks:

- suggest the career path
- justify the recommendation

Output format:

Career recommendation:

Reasons:

Benefits:

## Where This Prompt Type Is Used

- AI decision support systems
- consulting tools
- career guidance platforms
- recommendation systems

### Your Practice Task

Improve this prompt:

Suggest a business idea.

Try to include:

- role
- user goal
- justification requirement
- output format

## Exercise 57: Prompt for Critical Thinking

### Core Idea

Critical thinking prompts ask AI to **analyze an idea deeply instead of accepting it blindly**.

AI evaluates:

- strengths

- weaknesses
- assumptions
- possible improvements

This is useful for:

- research analysis
- business evaluation
- decision making

### **Example 1: Weak Prompt**

Is this idea good?

Problem:

- Only simple answer
- No deep analysis

### **Improved Prompt**

You are a critical thinking analyst.

Analyze the following business idea:

Idea: An AI-based learning platform for students.

Tasks:

1. Identify strengths of the idea
2. Identify weaknesses or risks

### 3. Suggest improvements

Output format:

Strengths:

Weaknesses:

Improvement suggestions:

### Why This Prompt Works

It forces AI to **evaluate the idea objectively** instead of giving only positive feedback.

### Real-Life Example

Prompt:

You are a research analyst.

Critically evaluate the idea of using AI in education.

Tasks:

- identify benefits
- identify risks
- suggest improvements

Output format:

Benefits:

Risks:

Improvements:

### **Where This Prompt Type Is Used**

- research analysis
- product evaluation
- policy analysis
- strategy planning

### **Your Practice Task**

Improve this prompt:

Analyze this idea.

Try to include:

- role
- evaluation tasks
- output format

## Exercise 58: Prompt for Risk Analysis

### Core Idea

Risk analysis prompts help AI **identify potential risks before starting a project or decision.**

AI can analyze risks such as:

- financial risk

- technical risk
- market risk
- operational risk

This is useful for:

- startup planning
- investment decisions
- project management
- business strategy

### **Example 1: Weak Prompt**

What are the risks?

Problem:

- No context
- No structured analysis

### **Improved Prompt**

You are a business risk analyst.

Analyze the risks of starting an **online AI education platform**.

Tasks:

1. Identify financial risks
2. Identify market risks
3. Identify technical risks
4. Suggest ways to reduce these risks

Output format:

Financial risks:

Market risks:

Technical risks:

Risk mitigation strategies:

### **Why This Prompt Works**

It forces AI to **analyze different types of risks systematically**.

### **Real-Life Example**

Prompt:

You are a project risk consultant.

Analyze risks involved in launching a new mobile application.

Tasks:



- identify major risks
- suggest prevention strategies

Output format:

Risk category:

Risk description:

Prevention strategy:

### **Where This Prompt Type Is Used**

- startup evaluation
- investment planning
- product development
- project risk management

### **Your Practice Task**

Improve this prompt:

Identify risks in this project.

Try to include:

- role
- project context
- risk categories
- output format

## Exercise 59: Prompt for Improvement Suggestions

### Core Idea

Improvement prompts help AI **analyze something and suggest ways to improve it.**

AI can improve:

- products

- business ideas
- learning methods
- websites
- processes

This is useful for **optimization and performance improvement**.

### **Example 1: Weak Prompt**

How can I improve this?

Problem:

- No context
- No evaluation criteria

### **Improved Prompt**

You are a product improvement consultant.

Analyze the following product idea:

Product: Online coding course for beginners.

Tasks:

1. Identify current limitations
2. Suggest improvement ideas
3. Recommend additional features

Output format:

Current limitations:

Improvement ideas:

Additional features:

### **Why This Prompt Works**

It guides AI to:

- identify weaknesses
- suggest improvements
- propose new features

This produces **actionable improvement suggestions**.

### **Real-Life Example**

Prompt:

You are a website optimization expert.

Analyze an online learning website.

Tasks:

- identify usability issues
- suggest improvements

Output format:

Issues found:

Improvement suggestions:

### **Where This Prompt Type Is Used**

- product optimization
- website improvement
- business process improvement
- service enhancement

### **Your Practice Task**

Improve this prompt:

Improve this system.

Try to include:

- role
- system context
- improvement tasks
- output format

## Exercise 60: Prompt for Strategic Planning

### Core Idea

Strategic planning prompts help AI **create long-term plans to achieve major goals.**

Unlike simple advice, strategy focuses on:

- long-term vision

- major actions
- resource planning
- growth steps

This is useful for:

- business growth
- career development
- startup planning
- organizational leadership

### **Example 1: Weak Prompt**

How can I grow my company?

Problem:

- No industry context
- No timeline
- No structured strategy

### **Improved Prompt**

You are a business strategy consultant.

Create a **3-year growth strategy** for an online programming education company.

Tasks:

1. Identify growth opportunities
2. Suggest strategic initiatives
3. Recommend marketing strategies
4. Suggest key milestones

Output format:

Growth opportunities:

Strategic initiatives:

Marketing strategy:

Key milestones:

### **Why This Prompt Works**

It focuses on **long-term planning instead of random suggestions**.

### **Real-Life Example**

Prompt:

You are a career strategist.



Create a **5-year career growth strategy** for someone who wants to become an AI engineer.

Tasks:

- identify skills to learn
- suggest career milestones

Output format:

Skill development:

Career milestones:

Growth strategy:

### **Where This Prompt Type Is Used**

- leadership planning
- startup strategy
- career growth planning
- organizational development

### **Your Practice Task**

Improve this prompt:

Plan my future career.

Try to include:

- role
- career goal
- planning timeline
- output format

## Exercise 61: Prompt for Teaching Through Examples

### Core Idea

Many people learn better through **examples instead of theory**.

Prompt engineers can ask AI to **teach concepts using examples first**.

This is useful for:

- programming education
- technical learning
- beginner training
- complex concept understanding

### **Example 1: Weak Prompt**

Explain machine learning.

Problem:

- AI may give theoretical explanation
- Beginners may not understand

### **Improved Prompt**

You are an AI instructor.

Teach the concept of **machine learning using examples**.

Tasks:

1. Give a simple real-life example
2. Explain the concept using that example
3. Provide one technical example

Output format:

Real-life example:

Concept explanation:

Technical example:

### Why This Prompt Works

It teaches using **example → explanation → technical understanding**.

This improves **learning clarity**.

### Real-Life Example

Prompt:

You are a Python teacher.

Teach the concept of **loops in Python** using examples.

Tasks:

- show a real-life analogy
- show Python code example

Output format:

Real-life analogy:

Code example:

Explanation:

### **Where This Prompt Type Is Used**

- AI tutors
- programming education
- training platforms
- educational content creation

### **Your Practice Task**

Improve this prompt:

Teach me about databases.

Try to include:

- role
- example-based teaching
- output format

## Exercise 62: Prompt for Knowledge Comparison

### Core Idea

Sometimes learners understand a topic better when they **compare two related concepts**.

Prompt engineers use comparison prompts to:

- highlight differences

- show similarities
- clarify confusion between concepts

This is useful for:

- education
- technical learning
- concept clarification

### **Example 1: Weak Prompt**

Explain AI and machine learning.

Problem:

- Topics explained separately
- No comparison

### **Improved Prompt**

You are a technology instructor.

Compare **Artificial Intelligence and Machine Learning**.

Tasks:

1. Explain each concept briefly
2. Identify similarities
3. Identify key differences

Output format:

Artificial Intelligence:

Machine Learning:

Similarities:

Differences:

### Why This Prompt Works

It helps learners understand:

- how concepts relate
- how they differ

This improves **concept clarity**.

### Real-Life Example

Prompt:

You are a programming mentor.

Compare **Python and JavaScript** for beginners.

Tasks:

- explain use cases
- highlight differences

Output format:



Python:

JavaScript:

Similarities:

Differences:

### **Where This Prompt Type Is Used**

- technical education
- concept clarification
- training platforms
- interview preparation

### **Your Practice Task**

Improve this prompt:

Compare two technologies.

Try to include:

- role
- specific technologies
- comparison tasks

- output format

## Exercise 63: Prompt for Knowledge Transformation

### Core Idea

Knowledge transformation prompts ask AI to **convert information into a different format**.

Examples:

- text → bullet points

- article → summary
- notes → quiz questions
- paragraph → table

This is useful for:

- education
- documentation
- productivity tools
- content repurposing

### **Example 1: Weak Prompt**

Convert this text.

Problem:

- No target format
- AI may guess the output

### **Improved Prompt**

You are a study assistant.

Convert the following article into structured study notes.

Tasks:

1. Extract key points
2. Convert them into bullet points
3. Add a short summary

Output format:

Key points:

- 
- 
- 

Summary:

## Why This Prompt Works

It clearly defines:

- role → study assistant
- transformation → article → notes
- structured output

This produces **clean study material**.

## Real-Life Example

Prompt:

You are a content editor.

Convert the following blog post into a **table format summary**.

Output format:

| Topic | Key Idea |

## Where This Prompt Type Is Used

- study note generators
- content repurposing tools
- knowledge management systems
- productivity tools

## Your Practice Task

Improve this prompt:

Convert this content into notes.

Try to include:

- role
- transformation task
- output format

## Exercise 64: Prompt for Knowledge Compression

### Core Idea

Knowledge compression means **reducing large information into a very small form without losing meaning.**

AI can compress content into:

- short summaries

- key takeaways
- quick notes

This is useful for:

- exam preparation
- quick revision
- executive summaries

### **Example 1: Weak Prompt**

Summarize this text.

Problem:

- No compression target
- AI may give long summary

### **Improved Prompt**

You are a study summarization expert.

Compress the following content about Artificial Intelligence.

Constraints:

- Maximum 5 key points
- Each point under 10 words

Output format:

Key Points:

- 1.
- 2.
- 3.
- 4.
- 5.

### Why This Prompt Works

It forces AI to **compress information into concise insights**.

This is very useful for **quick learning**.

### Real-Life Example

Prompt:

You are a business analyst.

Summarize the following report into **3 executive insights**.

Constraints:

- Each insight under 15 words

Output format:



Insight 1:

Insight 2:

Insight 3:

### **Where This Prompt Type Is Used**

- executive summaries
- exam revision notes
- research highlights
- quick knowledge review

### **Your Practice Task**

Improve this prompt:

Summarize this information.

Try to include:

- role
- compression limit

- output format

## Exercise 65: Prompt for Teaching with Analogies

### Core Idea

Analogies help explain **complex topics using familiar examples**.

AI can use everyday examples like:

- traffic systems

- school classrooms
- cooking recipes
- sports teams

This makes difficult topics **easy to understand**.

### **Example 1: Weak Prompt**

Explain blockchain.

Problem:

- Explanation may be technical
- Beginners may struggle

### **Improved Prompt**

You are a friendly teacher.

Explain **blockchain technology** using a simple analogy.

Tasks:

1. Give a daily-life analogy
2. Explain the concept using that analogy
3. Provide a short technical explanation

Output format:

Analogy:

Explanation using analogy:

Technical explanation:

### Why This Prompt Works

It teaches concepts using **familiar experiences**, making learning easier.

### Real-Life Example

Prompt:

You are a computer science teacher.

Explain **computer networking** using a real-life analogy.

Output format:

Analogy:

Concept explanation:

Summary:

### Where This Prompt Type Is Used

- education platforms
- beginner training

- concept explanation tools
- AI tutors

## Your Practice Task

Improve this prompt:

Explain machine learning.

Try to include:

- role
- analogy-based teaching
- output format

## Exercise 66: Prompt for Guided Learning

### Core Idea

Guided learning prompts make AI **teach step-by-step while interacting with the learner.**

Instead of giving everything at once, AI:

1. explains a small concept

2. asks a question
3. waits for the learner's answer
4. gives feedback

This creates **interactive learning**.

### **Example 1: Weak Prompt**

Teach me Python.

Problem:

- AI may give long explanation
- No interaction

### **Improved Prompt**

You are a Python tutor.

Teach Python step-by-step.

Process:

1. Explain a small concept
2. Ask one practice question
3. Wait for my answer

#### 4. Give feedback and explanation

Output format:

Concept explanation:

Practice question:

(Wait for student answer)

Feedback:

#### **Why This Prompt Works**

It creates **interactive learning instead of passive reading**.

#### **Real-Life Example**

Prompt:

You are a mathematics tutor.

Teach algebra step-by-step.

Process:

- explain concept
- ask a problem

- evaluate the answer

Output format:

Explanation:

Question:

Evaluation:

### **Where This Prompt Type Is Used**

- AI tutoring systems
- interactive learning platforms
- coding practice tools
- exam preparation systems

### **Your Practice Task**

Improve this prompt:

Teach me programming.

Try to include:

- role
- guided learning process



- interaction steps
- output format

## Exercise 67: Prompt for Knowledge Gap Detection

### Core Idea

Knowledge gap prompts help AI **identify what a learner does NOT know yet.**

AI analyzes:

- current knowledge
- missing concepts
- areas needing improvement

This is very useful for:

- education platforms
- skill assessment tools
- career learning plans

### **Example 1: Weak Prompt**

What should I learn?

Problem:

- AI does not know your current level
- Advice becomes generic

### **Improved Prompt**

You are a learning assessment mentor.

Evaluate the following learner profile.

Profile:

- Knows Python basics
- Understands variables and loops

- Does not know data structures or algorithms

Tasks:

1. Identify knowledge gaps
2. Suggest topics to learn next
3. Provide learning priority

Output format:

Knowledge gaps:

Next topics to learn:

Learning priority:

## Why This Prompt Works

It helps AI **diagnose missing knowledge and recommend learning steps.**

## Real-Life Example

Prompt:

You are a programming mentor.

Evaluate a student learning data science.

Known skills:

- Python basics

- basic statistics

Tasks:

- identify missing skills
- suggest next learning topics

Output format:

Missing skills:

Next topics:

Learning plan:

### **Where This Prompt Type Is Used**

- AI learning platforms
- career development tools
- skill assessment systems
- coaching programs

### **Your Practice Task**

Improve this prompt:

Tell me what I should learn next.

Try to include:

- role
- learner profile

- analysis tasks
- output format

## Exercise 68: Prompt for Process Optimization

### Core Idea

Process optimization prompts help AI **analyze a process and suggest ways to make it faster or better.**

AI can optimize:

- business workflows
- study methods
- customer service processes
- operational systems

This helps improve:

- efficiency
- productivity
- cost reduction

### **Example 1: Weak Prompt**

Improve this process.

Problem:

- No process details
- No optimization criteria

## Improved Prompt

You are a process optimization expert.

Analyze the following workflow for an online course platform.

Current process:

1. Student registers on website
2. Manual verification by staff
3. Course access given by email
4. Student joins the course

Tasks:

1. Identify inefficiencies
2. Suggest improvements
3. Recommend automation opportunities

Output format:

Current issues:

Improvement suggestions:

Automation opportunities:

## Why This Prompt Works

It guides AI to **analyze the process and propose improvements systematically.**

## Real-Life Example

Prompt:

You are an operations consultant.

Analyze the process of managing student enrollments in an online learning platform.

Tasks:

- identify delays
- suggest improvements

Output format:

Process issues:

Improvement suggestions:

## Where This Prompt Type Is Used

- business operations improvement
- workflow automation
- productivity optimization
- service process improvement

## Your Practice Task



Improve this prompt:

Improve my workflow.

Try to include:

- role
- workflow description
- optimization tasks
- output format

## Exercise 69: Prompt for Systems Thinking

### Core Idea

Systems thinking means **analyzing how different parts of a system interact with each other.**

Instead of looking at one problem, AI examines:

- components of the system
- relationships between components
- overall system behavior

This is useful for:

- business systems
- technology systems
- organizational planning
- complex problem solving

### **Example 1: Weak Prompt**

Analyze this business.

Problem:

- Only surface-level analysis
- No system perspective

### **Improved Prompt**

You are a systems thinking consultant.

Analyze an **online education business** as a system.

Tasks:

1. Identify key components of the system
2. Explain relationships between components

### 3. Identify potential system bottlenecks

Output format:

System components:

Relationships between components:

Potential bottlenecks:

### Why This Prompt Works

It helps AI understand **how the entire system works**, not just individual parts.

### Real-Life Example

Prompt:

You are a business systems analyst.

Analyze the system of running an online coding academy.

Tasks:

- identify key components (students, instructors, platform, marketing)
- explain how they interact

Output format:

System components:

Interactions:

Possible bottlenecks:

### **Where This Prompt Type Is Used**

- business system design
- organizational analysis
- technology architecture
- complex system evaluation

### **Your Practice Task**

Improve this prompt:

Analyze this system.

Try to include:

- role
- system context
- analysis tasks

- output format

## Exercise 70: Prompt for Root Cause Analysis

### Core Idea

Root cause analysis prompts help AI **find the real reason behind a problem.**

Instead of fixing symptoms, AI tries to identify:

- underlying causes

- contributing factors
- corrective actions

This is widely used in:

- business problem solving
- technical troubleshooting
- quality improvement

### **Example 1: Weak Prompt**

Why is my business not growing?

Problem:

- Too vague
- AI may give generic advice

### **Improved Prompt**

You are a business problem-solving consultant.

Analyze why an online course platform is not growing.

Tasks:

1. Identify possible root causes
2. Explain contributing factors
3. Suggest corrective actions

Output format:

Possible root causes:

Contributing factors:

Corrective actions:

### **Why This Prompt Works**

It guides AI to **diagnose the real issue instead of giving surface-level suggestions.**

### **Real-Life Example**

Prompt:

You are a technical troubleshooting expert.

Analyze why a website is experiencing frequent downtime.

Tasks:

- identify root causes
- suggest fixes

Output format:

Root causes:

Explanation:

Recommended fixes:

### **Where This Prompt Type Is Used**

- business consulting
- IT troubleshooting
- operational improvement
- quality management

### **Your Practice Task**

Improve this prompt:

Find the reason for this problem.

Try to include:

- role
- problem context
- analysis tasks
- output format



## Exercise 71: Prompt for Solution Generation

### Core Idea

Solution generation prompts help AI **create multiple possible solutions for a problem.**

Instead of giving one answer, AI can:

- generate several solutions

- evaluate each solution
- suggest the best option

This is useful for:

- innovation
- problem solving
- business strategy
- product development

### **Example 1: Weak Prompt**

How can I solve this problem?

Problem:

- No context
- No structured solutions

### **Improved Prompt**

You are a problem-solving expert.

Generate solutions for the following problem.

Problem:

Students find it difficult to stay consistent while learning programming.

Tasks:

1. Generate 5 possible solutions
2. Explain how each solution works
3. Suggest the most effective solution

Output format:

Solution 1:

Explanation:

Solution 2:

Explanation:

Solution 3:

Explanation:

Best solution

## Why This Prompt Works

It helps AI **explore multiple approaches instead of giving one quick answer.**

## Real-Life Example

Prompt:

You are an education innovation consultant.

Suggest solutions to improve student engagement in online courses.

Tasks:

- generate 5 ideas
- evaluate their effectiveness

Output format:

Idea:

Explanation:

Expected impact:

### **Where This Prompt Type Is Used**

- innovation brainstorming
- business consulting
- product development
- education improvement

### **Your Practice Task**

Improve this prompt:

Suggest solutions for my problem.

Try to include:

- role

- problem context
- number of solutions
- output format

## Exercise 72: Prompt for Structured Brainstorming

### Core Idea

Brainstorming prompts help generate **many ideas quickly**.  
But structured brainstorming organizes ideas into **categories**.  
This helps in:

- product development
- marketing campaigns
- startup idea generation
- innovation workshops

### **Example 1: Weak Prompt**

Give marketing ideas.

Problem:

- Random ideas
- No structure

### **Improved Prompt**

You are a marketing strategist.

Generate creative marketing ideas for promoting an online Python programming course.

Tasks:

1. Suggest social media marketing ideas
2. Suggest content marketing ideas

### 3. Suggest community engagement ideas

Output format:

Social media ideas:

Content marketing ideas:

Community engagement ideas:

### Why This Prompt Works

It organizes brainstorming into **clear categories**, making ideas easier to use.

### Real-Life Example

Prompt:

You are a startup innovation consultant.

Generate business ideas related to Artificial Intelligence.

Categories:

- education
- healthcare
- finance

Output format:

Education ideas:

Healthcare ideas:

Finance ideas:

## Where This Prompt Type Is Used

- startup brainstorming
- marketing planning
- innovation workshops
- product ideation

## Your Practice Task

Improve this prompt:

Give ideas for a new product.

Try to include:

- role
- product category
- idea categories
- output format

## Exercise 73: Prompt for Concept Breakdown

### Core Idea

Concept breakdown prompts help AI **divide a complex concept into smaller understandable parts.**



Instead of explaining everything together, AI:

1. identifies the main concept
2. breaks it into sub-concepts
3. explains each part

This helps learners **understand difficult topics step-by-step**.

### **Example 1: Weak Prompt**

Explain deep learning.

Problem:

- AI may give a complicated explanation
- Beginners may feel confused

### **Improved Prompt**

You are a machine learning instructor.

Break down the concept of **deep learning** into simpler subtopics.

Tasks:

1. Identify the main components of deep learning

2. Explain each component in simple language

3. Show how the components connect

Output format:

Main concept:

Subtopic 1:

Explanation:

Subtopic 2:

Explanation:

Subtopic 3:

Explanation:

Connection between components:

## Why This Prompt Works

It helps learners **understand the structure of a concept instead of memorizing definitions.**

## Real-Life Example

Prompt:

You are a programming mentor.

Break down the concept of **web development**.

Tasks:

- identify core components (frontend, backend, database)
- explain each component

Output format:

Frontend:

Explanation:

Backend:

Explanation:

Database:

Explanation:

### **Where This Prompt Type Is Used**

- AI tutoring systems
- educational platforms
- technical training
- concept explanation tools

### **Your Practice Task**

Improve this prompt:

Explain cloud computing.

Try to include:

- role

- concept breakdown tasks
- output format

## **Exercise 74: Prompt for Process Explanation**

### **Core Idea**

Process explanation prompts help AI **explain how something works step-by-step**.

Instead of giving a simple definition, AI explains:

- stages of the process
- actions at each stage
- final outcome

This is useful for:

- learning technical workflows
- understanding systems
- training beginners

### **Example 1: Weak Prompt**

Explain how machine learning works.

Problem:

- AI may give theoretical explanation
- Process may not be clear

### **Improved Prompt**

You are a machine learning teacher.

Explain the **process of how a machine learning model is built**.

Tasks:

1. Describe each stage of the process
2. Explain what happens in each stage
3. Give a simple example

Output format:

Step 1:

Explanation:

Step 2:

Explanation:

Step 3:

Explanation:

Example:

## Why This Prompt Works

It focuses on **how the process flows**, making it easier to understand.

## Real-Life Example

Prompt:

You are a software instructor.

Explain the **process of building a website**.

Tasks:

- identify stages
- explain each stage

Output format:

Planning stage:

Development stage:

Testing stage:

Deployment stage:

### **Where This Prompt Type Is Used**

- technical training
- software learning
- process documentation
- educational platforms

### **Your Practice Task**

Improve this prompt:

Explain how a system works.

Try to include:

- role
- system context
- process explanation steps
- output format

## **Exercise 75: Prompt for Scenario-Based Learning**

**Core Idea**



Scenario-based prompts teach through **real-life situations**.

Instead of only theory, AI creates a **practical situation** and explains what to do.

This helps learners:

- apply knowledge
- think practically
- solve real problems

### **Example 1: Weak Prompt**

Teach me cybersecurity.

Problem:

- Only theory
- No real-world application

### **Improved Prompt**

You are a cybersecurity instructor.

Teach cybersecurity using a real-world scenario.

Scenario:

A company employee receives a suspicious email asking for login details.

Tasks:

1. Explain what the problem is
2. Explain how the employee should respond
3. Explain how the company can prevent such attacks

Output format:

Scenario description:

Problem explanation:

Correct action:

Prevention strategies:

## Why This Prompt Works

It teaches through **real situations**, which improves understanding.

## Real-Life Example

Prompt:

You are a programming mentor.

Teach debugging using a scenario.

Scenario:

A Python program crashes with an error.

Tasks:

- explain how to analyze the error
- explain how to fix it

Output format:

Scenario:

Analysis:

Solution:

### **Where This Prompt Type Is Used**

- professional training
- technical education
- corporate learning
- AI tutoring systems

### **Your Practice Task**

Improve this prompt:

Teach me about software development.

Try to include:

- role
- scenario-based learning
- tasks
- output format

## **Exercise 76: Prompt for Framework Creation**

## Core Idea

Framework prompts help AI **create structured models for thinking and problem solving**.

A framework organizes ideas into a **clear structure**.

Examples of frameworks:

- SWOT analysis
- learning frameworks
- decision frameworks
- business models

This helps people **analyze problems systematically**.

## Example 1: Weak Prompt

Give tips to grow a business.

Problem:

- Random suggestions
- No structure

## Improved Prompt

You are a business strategist.

Create a framework for growing an online education business.

Tasks:

1. Identify key growth areas
2. Organize them into a structured framework
3. Explain each element

Output format:

Framework name:

Component 1:

Explanation:

Component 2:

Explanation:

Component 3:

Explanation:

## **Why This Prompt Works**

It creates a **structured thinking model instead of random advice.**

## **Real-Life Example**

Prompt:

You are a career development coach.

Create a framework for learning programming effectively.

Tasks:

- identify learning stages
- explain each stage

Output format:

Stage 1:

Explanation:

Stage 2:

Explanation:

Stage 3:

Explanation:

### **Where This Prompt Type Is Used**

- consulting
- strategy development
- education frameworks
- leadership training

### **Your Practice Task**

Improve this prompt:

Create a framework for success.

Try to include:

- role
- framework topic
- framework components
- output format

## **Exercise 77: Prompt for Concept Visualization**



## Core Idea

Concept visualization prompts help AI **explain ideas using visual structures**.

AI can present concepts using:

- diagrams
- flowcharts
- tables
- structured layouts

This helps people **see relationships between ideas**.

### Example 1: Weak Prompt

Explain data science.

Problem:

- Only text explanation
- Hard to visualize relationships

### Improved Prompt

You are a data science instructor.

Explain **data science as a visual concept structure**.

Tasks:

1. Identify main components of data science
2. Show relationships between components
3. Present the explanation in a diagram-like format

Output format:

Main concept: Data Science

Components:

| Component | Description |

Flow of process:

Step 1 → Step 2 → Step 3 → Step 4

## Why This Prompt Works

It converts abstract ideas into **visual structures**, which improves understanding.

## Real-Life Example

Prompt:

You are a programming teacher.

Explain the **software development lifecycle** using a flowchart-style explanation.

Output format:

Planning → Design → Development → Testing → Deployment

Each stage explanation.

### Where This Prompt Type Is Used

- technical education
- system design explanation
- process visualization
- training materials

### Your Practice Task

Improve this prompt:

Show how a system works.

Try to include:

- role
- system context
- visualization format
- output format

## **Exercise 78: Prompt for Learning Reinforcement**

## Core Idea

Learning reinforcement prompts help **strengthen memory after learning a topic.**

AI can reinforce learning by:

- reviewing key ideas
- asking questions
- providing small exercises

This helps learners **retain knowledge longer.**

## Example 1: Weak Prompt

Test me on Python.

Problem:

- No reinforcement structure
- Random questions

## Improved Prompt

You are a programming tutor.

Help reinforce my learning of **Python basics.**

Tasks:

1. Review 3 important concepts
2. Ask 2 practice questions
3. Provide feedback after my answer

Output format:

Concept review:

Practice question 1:

Practice question 2:

Feedback:

### Why This Prompt Works

It creates a **review + practice + feedback cycle**, which strengthens learning.

### Real-Life Example

Prompt:

You are an AI tutor.

Help reinforce my understanding of **machine learning basics**.

Tasks:

- review key ideas
- ask quiz questions

Output format:

Key concepts review:

Quiz question:

Explanation:

### **Where This Prompt Type Is Used**

- AI tutors
- exam preparation tools
- learning platforms
- skill training programs

### **Your Practice Task**

Improve this prompt:

Help me remember what I learned.

Try to include:

- role
- reinforcement tasks
- output format

## **Exercise 79: Prompt for Knowledge Expansion Questions**



## Core Idea

Sometimes after learning a topic, we should ask **deeper questions**.

Prompt engineers can ask AI to generate **thought-provoking questions** that expand knowledge.

These questions help learners:

- explore deeper concepts
- develop curiosity
- think critically

This is useful for:

- research learning
- advanced study
- innovation thinking

## Example 1: Weak Prompt

Ask questions about AI.

Problem:

- Questions may be basic

- No focus on deeper thinking

## Improved Prompt

You are a research mentor.

Generate **thought-provoking questions** about Artificial Intelligence.

Tasks:

1. Ask questions that encourage deeper understanding
2. Focus on future developments and ethical issues
3. Encourage critical thinking

Output format:

Question 1:

Question 2:

Question 3:

Question 4:

Question 5:

## Why This Prompt Works

It encourages **curiosity-driven learning**, which leads to deeper understanding.

## Real-Life Example

Prompt:

You are a science teacher.

Generate **advanced discussion questions** about machine learning.

Tasks:

- encourage critical thinking
- encourage research exploration

Output format:

Discussion question 1:

Discussion question 2:

Discussion question 3:

## Where This Prompt Type Is Used

- research training
- advanced education
- discussion-based learning
- innovation workshops

## **Your Practice Task**

Improve this prompt:

Ask questions about technology.

Try to include:

- role
- topic focus
- question type
- output format

## **Exercise 80: Prompt for Reflection**

### **Core Idea**

Reflection prompts help people **think about their experiences and learn from them.**

AI can guide reflection by asking questions like:

- What did you learn?
- What could be improved?
- What will you do differently next time?

This is useful for:

- personal growth
- leadership development
- learning improvement

### **Example 1: Weak Prompt**

Reflect on your work.

Problem:

- Too vague
- No structure

### **Improved Prompt**

You are a learning coach.

Guide a student to reflect on their learning experience in a programming course.

Tasks:

1. Ask questions about what they learned
2. Ask about challenges faced
3. Ask about improvements for future learning

Output format:

Learning reflection questions:

Question 1:

Question 2:

Question 3:

### **Why This Prompt Works**

It encourages **self-awareness and improvement**.

### **Real-Life Example**

Prompt:

You are a leadership coach.

Guide a manager to reflect on their team leadership experience.

Tasks:

- ask reflective questions
- encourage improvement thinking

Output format:

Reflection questions:

Improvement suggestions:

### **Where This Prompt Type Is Used**

- coaching programs
- leadership training
- personal development tools
- learning platforms

### **Your Practice Task**

Improve this prompt:

Help me reflect on my progress.

Try to include:

- role
- reflection focus
- tasks
- output format

## **Exercise 81: Prompt for Knowledge Application**

**Core Idea**



Learning is complete only when we **apply knowledge to real situations.**

Knowledge application prompts help AI:

- create practical tasks
- simulate real-world problems
- guide users to apply concepts

This is useful for:

- skill development
- professional training
- technical education

### **Example 1: Weak Prompt**

Give exercises for Python.

Problem:

- Random exercises
- No real-world context

### **Improved Prompt**

You are a Python programming instructor.

Create practical exercises to apply knowledge of **Python loops.**

Tasks:

1. Create a simple coding task
2. Create a real-world application task
3. Provide expected output

Output format:

Exercise 1:

Task description:

Expected output:

Exercise 2:

Task description:

Expected output:

## Why This Prompt Works

It encourages **learning by doing**, which strengthens understanding.

## Real-Life Example

Prompt:

You are a data science instructor.

Create a practical exercise using **data analysis in Python**.

Tasks:

- provide dataset example
- provide analysis task

Output format:

Dataset description:

Task:

Expected result:

### **Where This Prompt Type Is Used**

- coding practice platforms
- professional training programs
- AI learning tools
- skill development courses

### **Your Practice Task**

Improve this prompt:

Give practice exercises.

Try to include:

- role
- skill/topic
- application tasks
- output format

## **Exercise 82: Prompt for Adaptive Learning**

## Core Idea

Adaptive learning prompts help AI **adjust teaching based on the learner's level.**

AI first understands:

- current skill level
- learning goals
- difficulties

Then AI **changes the teaching style or difficulty level.**

This is useful for:

- AI tutors
- personalized learning platforms
- skill development programs

## Example 1: Weak Prompt

Teach me programming.

Problem:

- AI does not know your level
- Explanation may be too hard or too easy

## Improved Prompt

You are a programming tutor.

Teach programming based on the learner's level.

Learner profile:

Skill level: Beginner

Goal: Learn Python for data science

Tasks:

1. Adjust explanation for beginner level
2. Provide simple examples
3. Suggest next learning step

Output format:

Concept explanation:

Example:

Next step:

## Why This Prompt Works

It allows AI to **adapt the teaching style to the learner's needs.**

## Real-Life Example

Prompt:

You are a mathematics tutor.

Teach algebra based on student level.

Student level: Intermediate

Tasks:

- adjust difficulty level
- give appropriate problems

Output format:

Explanation:

Practice problem:

Solution:

## Where This Prompt Type Is Used

- AI tutoring systems
- online learning platforms
- skill training tools
- education technology

## Your Practice Task

Improve this prompt:

Teach me a skill.

Try to include:

- role
- learner profile
- adaptive teaching tasks
- output format

## **Exercise 83: Prompt for Knowledge Synthesis**



## Core Idea

Knowledge synthesis means **combining multiple pieces of information to form a clear conclusion or insight.**

Instead of analyzing each idea separately, AI:

- collects related ideas
- connects them
- produces a final understanding

This is useful for:

- research summaries
- strategic insights
- learning synthesis

## Example 1: Weak Prompt

Explain these topics together.

Problem:

- AI may explain topics separately
- No final insight

## Improved Prompt

You are a research analyst.

Synthesize the following topics into a unified understanding.

Topics:

- Artificial Intelligence
- Machine Learning
- Data Science

Tasks:

1. Explain each topic briefly
2. Show how they work together
3. Provide a final insight

Output format:

Topic overview:

How they connect:

Final insight:

## Why This Prompt Works

It helps produce **integrated knowledge instead of isolated explanations.**

## Real-Life Example

Prompt:

You are a technology strategist.

Combine insights from the following areas:

- cloud computing
- artificial intelligence
- big data

Tasks:

- explain relationship
- give strategic insight

Output format:

Technology overview:

Relationship between technologies:

Strategic insight:

### **Where This Prompt Type Is Used**

- research analysis
- technology strategy
- interdisciplinary learning
- business intelligence

### **Your Practice Task**

Improve this prompt:

Combine these ideas.

Try to include:

- role
- topics
- synthesis tasks
- output format

## **Exercise 84: Prompt for Knowledge Translation**

## Core Idea

Knowledge translation means **converting complex technical information into simple language**.

AI can translate knowledge for different audiences like:

- beginners
- non-technical people
- school students
- business leaders

This is useful for:

- teaching
- technical communication
- documentation

## Example 1: Weak Prompt

Explain machine learning simply.

Problem:

- No target audience
- AI may still use technical terms

## Improved Prompt

You are a technology educator.

Translate the concept of **machine learning** into simple language for a non-technical audience.

Tasks:

1. Avoid technical jargon
2. Use simple examples
3. Provide a short summary

Output format:

Simple explanation:

Example:

Summary:

### Why This Prompt Works

It ensures the explanation becomes **accessible to people without technical background**.

### Real-Life Example

Prompt:

You are a science communicator.

Explain **cloud computing** to a school student.

Tasks:

- use simple language
- use a daily-life example

Output format:

Explanation:

Example:

Summary:

### **Where This Prompt Type Is Used**

- education platforms
- technical documentation
- science communication
- training materials

### **Your Practice Task**

Improve this prompt:

Explain artificial intelligence simply.

Try to include:

- role
- target audience
- translation tasks
- output format

## **Exercise 85: Prompt for Insight Extraction**



## Core Idea

Insight extraction prompts help AI **find meaningful conclusions from information or data**.

Instead of just repeating the information, AI:

- identifies key insights
- highlights important findings
- suggests implications

This is useful for:

- business intelligence
- research analysis
- strategic decisions

## Example 1: Weak Prompt

Analyze this report.

Problem:

- AI may summarize instead of finding insights

## Improved Prompt

You are a business intelligence analyst.

Extract key insights from the following sales report.

Tasks:

1. Identify major trends
2. Identify surprising findings
3. Suggest implications for business decisions

Output format:

Key insights:

Unexpected findings:

Business implications:

### **Why This Prompt Works**

It focuses on **discovering valuable insights instead of simple summarization.**

### **Real-Life Example**

Prompt:

You are a market research analyst.

Analyze customer feedback data.

Tasks:

- identify key customer concerns
- identify improvement opportunities

Output format:

Customer concerns:

Improvement opportunities:

### **Where This Prompt Type Is Used**

- market research
- business analytics
- product improvement
- strategic planning

### **Your Practice Task**

Improve this prompt:

Find important points in this data.

Try to include:

- role
- data context
- insight extraction tasks
- output format

## **Exercise 86: Prompt for Strategic Recommendations**

## Core Idea

Strategic recommendation prompts help AI **suggest high-level actions based on analysis.**

Instead of giving random advice, AI:

1. analyzes the situation
2. identifies opportunities
3. recommends strategic actions

This is useful for:

- business consulting
- marketing strategy
- product development
- leadership decisions

## Example 1: Weak Prompt

Give suggestions to improve my business.

Problem:

- No context
- Suggestions may be generic

## Improved Prompt

You are a business strategy consultant.

Provide strategic recommendations for improving an **online programming training platform**.

Tasks:

1. Identify growth opportunities
2. Suggest strategic initiatives
3. Recommend priority actions

Output format:

Growth opportunities:

Strategic initiatives:

Priority actions:

### **Why This Prompt Works**

It encourages **strategic thinking instead of simple tips**.

### **Real-Life Example**

Prompt:

You are a digital marketing strategist.

Recommend strategies to increase enrollment in an online Python course.

Tasks:

- identify target audience
- suggest marketing strategies

Output format:

Target audience:

Marketing strategies:

Expected impact:

### **Where This Prompt Type Is Used**

- consulting tools
- strategic planning systems
- marketing analysis
- business decision support

### **Your Practice Task**

Improve this prompt:

Suggest ways to grow my project.

Try to include:

- role
- project context
- recommendation tasks
- output format

## **Exercise 87: Prompt for Knowledge Distillation**



## Core Idea

Knowledge distillation means **extracting the core essence of a complex topic**.

Instead of long explanations, AI identifies:

- the most important idea
- supporting principles
- the core takeaway

This helps learners **grasp the essence quickly**.

## Example 1: Weak Prompt

Explain deep learning briefly.

Problem:

- AI may still give unnecessary details

## Improved Prompt

You are a technology educator.

Distill the concept of **deep learning** into its essential ideas.

Tasks:

1. Identify the core concept
2. Identify 3 supporting principles
3. Provide one key takeaway

Output format:

Core concept:

Supporting principles:

Key takeaway:

## Why This Prompt Works

It forces AI to **focus on the essence instead of details**.

## Real-Life Example

Prompt:

You are a learning coach.

Distill the concept of **software engineering** into the most important ideas.

Output format:

Core idea:

Key principles:

Main takeaway:

## **Where This Prompt Type Is Used**

- executive summaries
- learning frameworks
- teaching simplified concepts
- knowledge management

## **Your Practice Task**

Improve this prompt:

Explain the main idea of this topic.

Try to include:

- role
- distillation tasks
- output format

## **Exercise 88: Prompt for Knowledge Generalization**

## Core Idea

Knowledge generalization means **taking lessons from one example and applying them to broader situations.**

AI can:

- analyze a specific case
- extract general principles
- show how those principles apply in other contexts

This helps people **learn transferable knowledge.**

## Example 1: Weak Prompt

What can we learn from this example?

Problem:

- No structured learning extraction

## Improved Prompt

You are a learning analyst.

Analyze the following example and extract general lessons.

Example:

A startup succeeded by focusing on a single niche market before expanding.

Tasks:

1. Identify the key lesson from this example
2. Extract general principles
3. Show how these principles apply to other businesses

Output format:

Key lesson:

General principles:

Applications in other situations:

### **Why This Prompt Works**

It converts a **single example into universal learning principles**.

### **Real-Life Example**

Prompt:

You are a business mentor.

Analyze the success of a technology company.

Tasks:

- extract business lessons
- generalize them for startups

Output format:

Key lesson:

General principles:

Application for startups:

### **Where This Prompt Type Is Used**

- business case studies
- leadership learning
- strategic analysis
- educational frameworks

### **Your Practice Task**

Improve this prompt:

What can we learn from this situation?

Try to include:

- role
- example context
- generalization tasks
- output format

## **Exercise 89: Prompt for Meta-Learning**

## Core Idea

Meta-learning means **learning how to learn better**.

Instead of only learning a topic, AI helps analyze:

- learning methods
- study strategies
- improvement techniques

This helps people **learn faster and more effectively**.

## Example 1: Weak Prompt

How can I study better?

Problem:

- AI may give general advice
- No structured analysis

## Improved Prompt

You are a learning strategy expert.

Help a student improve their learning method.

Student profile:

Goal: Learn programming

Current issue: Difficulty remembering concepts

Tasks:

1. Identify possible problems in the study approach
2. Suggest better learning techniques
3. Recommend a study routine

Output format:

Learning problems:

Improved learning techniques:

Recommended study routine:

## Why This Prompt Works

It focuses on **improving the learning process**, not just the subject.

## Real-Life Example

Prompt:

You are an education coach.



Help improve the learning strategy of someone studying data science.

Tasks:

- analyze study habits
- suggest learning improvements

Output format:

Current issues:

Better strategies:

Learning plan:

### **Where This Prompt Type Is Used**

- AI tutoring systems
- study improvement tools
- coaching programs
- productivity learning

### **Your Practice Task**

Improve this prompt:

Help me learn faster.

Try to include:

- role
- learning goal
- analysis tasks
- output format

## Exercise 90: Prompt for Final Knowledge Integration

### Core Idea

After learning many concepts, we must **combine them into one structured system**.

Knowledge integration prompts help AI:

- organize multiple ideas
- connect them logically
- present a complete framework

This is useful for:

- course design
- research synthesis
- strategic thinking

### Example 1: Weak Prompt

Summarize everything we learned.

Problem:

- AI may produce random summary
- No structure

## Improved Prompt

You are a learning architect.

Integrate the following concepts into a structured learning system.

Concepts:

- Artificial Intelligence
- Machine Learning
- Data Science
- Programming

Tasks:

1. Explain how these topics connect
2. Organize them into a learning framework
3. Suggest learning sequence

Output format:

Concept relationships:

Learning framework:

Suggested learning sequence:

## Why This Prompt Works

It helps learners **see the complete system instead of isolated topics.**

## Real-Life Example

Prompt:

You are a curriculum designer.

Integrate the following topics into a course structure:

- Prompt Engineering
- AI Tools
- Automation
- AI Projects

Tasks:

- connect the topics
- design a learning flow

Output format:

Topic relationships:

Course framework:

Learning order:

## Where This Prompt Type Is Used

- course design
- curriculum development
- research frameworks
- knowledge management

### **Your Practice Task**

Improve this prompt:

Combine all the topics.

Try to include:

- role
- topics
- integration tasks
- output format

## Exercise 91: Prompt for AI System Design

### Core Idea

AI system design prompts help AI **design an entire AI-powered system or application**.

Instead of solving one task, AI plans:

- system components
- workflow
- user interaction

This is useful for:

- building AI tools
- designing automation systems
- product development

### Example 1: Weak Prompt

Create an AI app idea.

Problem:

- Only idea
- No system design

## Improved Prompt

You are an AI product architect.

Design an AI-powered **study assistant application**.

Tasks:

1. Identify core features
2. Design system components
3. Explain user workflow

Output format:

Core features:

System components:

User workflow:

## Why This Prompt Works

It helps create **complete system architecture instead of just ideas**.

## Real-Life Example



Prompt:

You are an AI product designer.

Design an AI-powered resume analyzer tool.

Tasks:

- define features
- define workflow

Output format:

Features:

System components:

Workflow:

### **Where This Prompt Type Is Used**

- AI product design
- software architecture
- automation systems
- AI tool development

## Your Practice Task

Improve this prompt:

Design an AI tool.

Try to include:

- role
- tool purpose
- design tasks
- output format

## Exercise 92: Prompt for Innovation Thinking

### Core Idea

Innovation prompts help AI **generate new and creative ideas to solve problems.**

Instead of normal solutions, AI is encouraged to:

- think differently
- combine ideas
- suggest novel approaches

This is useful for:

- startups
- product innovation
- creative problem solving
- research exploration

### Example 1: Weak Prompt

Give a new idea.

Problem:

- AI may give generic ideas
- No innovation guidance

## Improved Prompt

You are an innovation strategist.

Generate innovative ideas for improving online education.

Tasks:

1. Identify current limitations in online learning
2. Propose creative solutions
3. Suggest technologies that can enable these ideas

Output format:

Current limitations:

Innovative ideas:

Technologies to implement them:

## Why This Prompt Works

It encourages **creative thinking combined with practical implementation.**

## Real-Life Example

Prompt:

You are a startup innovation consultant.

Generate innovative product ideas using Artificial Intelligence in healthcare.

Tasks:

- identify healthcare problems
- propose AI-based solutions

Output format:

Healthcare problem:

AI solution idea:

Potential impact:

## Where This Prompt Type Is Used

- startup ideation
- research innovation
- product design
- technology development

## Your Practice Task

Improve this prompt:

Generate innovative ideas.

Try to include:

- role
- innovation area
- idea generation tasks
- output format

## Exercise 93: Prompt for AI Agent Design

### Core Idea

AI agent prompts help design **autonomous AI systems that perform tasks automatically.**

An AI agent can:

1. understand a goal
2. plan actions
3. execute tasks
4. observe results
5. improve decisions

This is useful for:

- automation systems
- productivity tools
- intelligent assistants

## Example 1: Weak Prompt

Build an AI assistant.

Problem:

- No defined tasks
- No workflow

## Improved Prompt

You are an AI systems architect.

Design an AI agent for helping students study programming.

Tasks:

1. Define the agent's goal
2. Define tasks the agent should perform
3. Explain how the agent interacts with users

Output format:

Agent goal:

Agent tasks:

User interaction workflow:

## Why This Prompt Works



It defines the **purpose, capabilities, and workflow of the AI agent.**

### **Real-Life Example**

Prompt:

You are an AI automation architect.

Design an AI agent that helps analyze resumes.

Tasks:

- define agent responsibilities
- define workflow

Output format:

Agent purpose:

Tasks performed:

Workflow:

### **Where This Prompt Type Is Used**

- AI assistants
- automation tools
- productivity agents
- intelligent systems

### **Your Practice Task**

Improve this prompt:

Create an AI assistant.

Try to include:

- role
- assistant purpose
- tasks
- output format

## Exercise 94: Prompt for Cross-Domain Thinking

### Core Idea

Cross-domain thinking means **combining ideas from different fields to create new insights**.

Innovation often happens when we connect knowledge from **two different domains**.

Examples:

- AI + healthcare
- psychology + marketing
- education + gaming
- finance + technology

This helps generate **powerful new ideas**.

### Example 1: Weak Prompt

Combine two ideas.

Problem:

- No domains defined
- No purpose

## Improved Prompt

You are an innovation strategist.

Combine ideas from **Artificial Intelligence and Education**.

Tasks:

1. Identify opportunities where AI can improve education
2. Suggest innovative solutions
3. Explain potential impact

Output format:

Opportunity area:

AI-based solution:

Potential impact:

## Why This Prompt Works

It forces AI to **merge knowledge from different domains**, producing creative insights.

## Real-Life Example

Prompt:

You are a product innovation consultant.

Combine ideas from **gaming and online learning**.

Tasks:

- identify ways to gamify education
- suggest practical features

Output format:

Learning challenge:

Gamified solution:

Expected benefit:

### **Where This Prompt Type Is Used**

- product innovation
- startup ideation
- interdisciplinary research
- creative problem solving

## Your Practice Task

Improve this prompt:

Combine ideas from different fields.

Try to include:

- role
- specific domains
- tasks
- output format

## Exercise 95: Prompt for Long-Term Vision Planning

### Core Idea

Long-term vision prompts help AI **design a future direction for a person, company, or project.**

Instead of short tips, AI focuses on:

- long-term goals
- major milestones
- growth strategy

This is useful for:

- leadership planning
- career vision
- startup growth
- strategic development

### Example 1: Weak Prompt

Plan my future.

Problem:

- No context
- No time horizon

## Improved Prompt

You are a strategic planning advisor.

Create a **10-year vision plan** for someone who wants to build a successful AI education company.

Tasks:

1. Define the long-term vision
2. Identify major milestones
3. Suggest strategic actions

Output format:

Long-term vision:

Major milestones:

Strategic actions:

## Why This Prompt Works

It focuses on **future direction and strategic milestones**.

## Real-Life Example



Prompt:

You are a career strategist.

Create a **10-year career vision plan** for someone who wants to become an AI expert.

Tasks:

- identify learning milestones
- suggest career stages

Output format:

Vision goal:

Learning milestones:

Career stages:

### Where This Prompt Type Is Used

- leadership coaching
- strategic planning
- career development
- long-term project planning

## Your Practice Task

Improve this prompt:

Plan my long-term goals.

Try to include:

- role
- goal context
- planning timeline
- output format

## Exercise 96: Prompt for Expert Simulation

### Core Idea

Expert simulation prompts make AI **behave like a domain expert**.

Instead of general answers, AI responds with **expert-level thinking**.

AI can simulate experts like:

- doctor
- business consultant
- software architect
- marketing strategist

This helps users **get expert-style guidance**.

### Example 1: Weak Prompt

Give advice about marketing.

Problem:

- Advice may be generic
- No expert perspective

## Improved Prompt

You are a digital marketing expert with 10 years of experience. Provide expert advice for promoting an online programming course.

Tasks:

1. Identify target audience
2. Suggest marketing channels
3. Recommend campaign strategies

Output format:

Target audience:

Marketing channels:

Campaign strategy:

## Why This Prompt Works

It makes AI **simulate an experienced professional**, leading to more realistic advice.

## Real-Life Example

Prompt:

You are a senior software architect.

Provide expert guidance for designing a scalable web application.

Tasks:

- identify architecture components
- suggest best practices

Output format:

Architecture components:

Best practices:

### **Where This Prompt Type Is Used**

- consulting tools
- professional guidance systems
- expert advisory platforms
- training simulations

### **Your Practice Task**

Improve this prompt:

Give professional advice.

Try to include:

- role
- domain expertise
- advice tasks
- output format

## Exercise 97: Prompt for AI Collaboration

### Core Idea

AI collaboration prompts treat AI as a **thinking partner instead of a simple tool**.

AI helps by:

- asking questions
- suggesting ideas
- improving your thinking

This creates **human + AI collaboration**.

### Example 1: Weak Prompt

Help me with my idea.

Problem:

- AI may give a single answer
- No collaboration

## Improved Prompt

You are my AI brainstorming partner.

Help me develop a new idea for an online learning platform.

Process:

1. Ask clarifying questions about my idea
2. Suggest improvements
3. Help refine the concept

Output format:

Clarifying questions:

Improvement suggestions:

Refined idea:

## Why This Prompt Works

It creates **interactive thinking between human and AI**.

## Real-Life Example



Prompt:

You are my startup brainstorming partner.

Help me refine a business idea related to Artificial Intelligence education.

Tasks:

- ask questions to understand the idea
- suggest improvements

Output format:

Questions:

Suggestions:

Refined concept:

### **Where This Prompt Type Is Used**

- creative brainstorming
- startup ideation
- collaborative thinking
- innovation planning

## Your Practice Task

Improve this prompt:

Help me think about this idea.

Try to include:

- role
- collaboration process
- tasks
- output format

## Exercise 98: Prompt for AI Self-Evaluation

### Core Idea

Self-evaluation prompts make AI **review its own answer and improve it.**

Normally AI gives one answer.

But with self-evaluation, AI will:

1. generate an answer
2. review the answer
3. improve the answer

This improves:

- accuracy
- clarity
- quality

### Example 1: Weak Prompt

Write an article about AI.

Problem:

- AI gives first draft only
- No quality check

## Improved Prompt

You are an AI writing assistant.

Write an article about **Artificial Intelligence in Education**.

Process:

1. Write the first draft
2. Review the draft for clarity and completeness
3. Improve the article

Output format:

First draft:

Self-review:

Improved version:

## Why This Prompt Works

It forces AI to **critically evaluate its own output**, producing a better final result.

## Real-Life Example

Prompt:

You are a content editor.

Write a blog post about **learning Python for beginners**.

Process:

- generate the first draft
- evaluate clarity and usefulness
- produce improved version

Output format:

Draft:

Review:

Final version:

### **Where This Prompt Type Is Used**

- writing tools
- research assistance
- coding assistants
- content creation systems

### **Your Practice Task**

Improve this prompt:

Improve your answer.

Try to include:

- role
- evaluation steps
- improvement process
- output format

## Exercise 99: Prompt for AI Reflection Loop

### Core Idea

Reflection loop prompts make AI **think, evaluate, and improve repeatedly**.

Instead of a single answer, AI follows a **thinking cycle**:

1. generate solution
2. check for mistakes
3. refine the solution

This technique is used in **advanced AI systems and reasoning models**.

### Example 1: Weak Prompt

Solve this problem.

Problem:

- AI gives one answer
- No improvement process

### Improved Prompt

You are an analytical problem-solving assistant.

Solve the following problem.

Process:

1. Generate an initial solution
2. Review the solution for errors or weaknesses
3. Improve the solution

Output format:

Initial solution:

Reflection on solution:

Improved solution:

## Why This Prompt Works

It creates a **thinking loop**, improving reasoning and accuracy.

## Real-Life Example

Prompt:

You are a coding assistant.

Write a Python program to calculate factorial.

Process:

- write initial code



- review the code for mistakes
- provide improved version

Output format:

Initial code:

Code review:

Improved code:

### **Where This Prompt Type Is Used**

- advanced reasoning systems
- coding assistants
- research analysis
- problem-solving tools

### **Your Practice Task**

Improve this prompt:

Think again and improve your answer.

Try to include:

- role
- reflection steps
- output format

## Exercise 100: Master Prompt Design

### Core Idea

A master prompt combines many techniques:

- role prompting
- structured tasks
- output formatting
- reasoning steps
- evaluation

This type of prompt is used to build **real AI tools and systems**.

### Example 1: Weak Prompt

Help me learn AI.

Problem:

- No structure
- No tasks
- No learning system

### Improved Master Prompt

You are an AI learning mentor.

Help a beginner learn **Artificial Intelligence step-by-step**.

Learner profile:

Level: Beginner

Goal: Become an AI engineer

Tasks:

1. Create a learning roadmap
2. Explain key concepts simply
3. Suggest practical exercises
4. Ask review questions

Output format:

Learning roadmap:

Concept explanation:

Practice exercise:

Review questions:

## Why This Prompt Works

It combines several prompt engineering techniques:

- role prompting
- structured learning
- practical application
- knowledge reinforcement

This is how **real AI-powered learning systems are built**.

### Real-Life Example

Prompt:

You are an AI career mentor.

Help a student become a **Data Scientist**.

Tasks:

1. Design a 6-month learning roadmap
2. Suggest projects to build
3. Recommend learning resources

Output format:

Learning roadmap:

Projects:

Resources:

### Where Master Prompts Are Used

- AI assistants
- education platforms
- automation tools
- AI-powered products

## **Your Final Practice Task**

Improve this prompt:

Help me achieve my goal.

Try to include:

- role
- goal context
- tasks
- output format